

CHAPTER XIII

QUMRAN'S PLASTERED POOLS: A NEW PERSPECTIVE

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The stepped and plastered pools¹ at Qumran are considered by many as the most important architectural element relating to the general interpretation of the site. Regardless of whether or not the pools are connected to purity and ritual immersion, they constitute an essential factor in determining the character of the settlement, the nature of its inhabitants, their religious affiliation and beliefs.

The highly varied and at times contradictory interpretations regarding the functions of these installations may be related to the desire to define a relationship between the archaeological material—the pools and the scrolls. Those who view the site and the scrolls as two separate, independent entities usually doubt the religious significance of the pools. But even those who view the texts and the site as interrelated do not seem to agree about which of the pools can be associated with the act of immersion, if at all, and which served other functions.

The intention of this chapter is not to undermine the historical importance of the scrolls. Whether or not these were copied at the site itself, and whether they can be related directly or indirectly to the inhabitants of Qumran, is not of relevance in this study, at least not for the analysis of the pools. Essentially, the scrolls come from the same cultural and geographical framework as the structural remains, and there clearly exists a chronological overlap between the scrolls and the site's main phases of occupation. This fact is in itself sufficient to view the scrolls as historical documents that can be of value for the interpretation of the material evidence. However, just as at any other site, regardless of the outstanding quality and quantity of the Dead Sea Scrolls, using the written evidence as a point of departure, or even using it to reconstruct the historical background of a given material culture, is contrary to archaeological methodology. Qumran should first be examined on its own merits without the support of the texts. Only once this process has been fully completed should one include the literary evidence, most importantly the Dead Sea Scrolls, Josephus and the Rabbinic literature, as a source of information that is solely supplementary or secondary to the material evidence. It is this methodology that has been the guideline for the present study of the pools.

In spite of the numerous studies that have addressed the issues of Qumran's stepped pools, very few seem to relate to the actual physical remains. Unlike de Vaux's interpretative methods and conclusions that have stimulated a variety of reactions among scholars and laymen, his descriptive and

analytical examination of the pools has hardly been questioned. His chronological reconstruction of the water system's development and the original top plans are considered by all as the ultimate reference. North (1962), Strobel (1972), Petit (1981), Wood (1984) and Hidirolou (2000), just to mention some of the more meticulous studies of Qumran's plastered installations, follow strictly de Vaux's original chronology, and merely reproduce the same original plans of these installations. The focus and only merit of these studies are their interpretations of the use and function of the pools, a topic that has the tolerance to accommodate a variety of speculative ideas.

Though the functional interpretation of the pools in the present study offers an alternative perspective, it is the newly verified data in the field and the newly drawn plans that should be considered as the most important contribution of this chapter. The recently published field notes of de Vaux (Humbert and Chambon 1996), as well as the Qumran-archives at the École Biblique et Archéologique Française (drawings, plans, photos and classified objects), made freely available to the author, constitute the basis for this study. The six-month permit of the Israel Antiquities Authority, July through December of 2000 has made possible the frequent visits to the site, a careful reexamination of the pools and the extraction of plaster samples. Father Jean-Baptiste Humbert, with the assistance of the author, has remeasured and newly recorded the pools. The original plans from the 1950s were copied and corrected where necessary, and sections were drawn for the first time. Without the generous financial support of the Biblical Archaeology Society this study would not have been possible.

MIQVEH OR STEPPED POOL—ETYMOLOGICAL CLARIFICATIONS

Some ten years prior to Yadin's identification of the ritual pools at Masada, de Vaux had already expressed his intuition that some of the Qumran pools served ritual purposes. It was, however, not until the 1960s that most stepped pools from the late Hellenistic, Roman and Byzantine periods in Palestine were designated as *miqva'ot*². Given the evolving and changing character of the word *miqveh*, this designation is problematic³.

Literally translated, the word *miqveh* means "collection," or "collected mass," especially of water. In the

biblical context (Gen 1: 10; Exod 7: 9; and Lev 11: 36) a *miqveh* most frequently evokes a natural body or source of water that has the quality to eliminate impurities from people or objects. It can, however, be used in a context that is not connected to purity or impurity. In Isaiah 22: 11, for example, the word *miqveh* designates a pond. “You made also a pond between the two walls for the water of the old pool.” It is only in the Rabbinic literature that a *miqveh* designates an artificial installation built for the purpose of ritual purity. However, it is clearly stated that the act of ritual immersion does not have to be the only act performed in this kind of installation. For example, rinsing food in a *miqveh* would not necessarily alter the ritual character of the water. In *Miqva’ot* 7: 3 it is stated that: “If one rinsed in it [an immersion-pool] baskets of olives and baskets of grapes, and they changed its colors, it is valid ... If wine and olive-sap fell into it and changed its color, it is unfit. What should one do? He should wait on it until rain falls and its color returns to the color of water.” Furthermore, most qualities and rules that apply for ritual immersion pools also apply to other bodies of water contained in natural depressions, containers or, alternatively, built installations such as cisterns and wells. In *Miqva’ot* 1: 4 it is clearly stated that “All the same are [or, the same rules apply for] (1) water of ponds, (2) water of cisterns, (3) water of ditches, (4) water of caverns, (5) water of rain-drippings which have stopped, and (6) immersion-pools which do not contain forty *seahs*: during the rainy season, all are clean.”

In medieval Europe, the *miqveh* represented one of the most important institutions within the Jewish community⁴. In urgent cases, it was even permitted to sell a synagogue in order to erect a *miqveh* (Berlin 1880, p. 45). Medieval *miqva’ot* were mainly built for the purpose of ritual immersion. However, in many instances they served as bathhouses because of the order forbidding Jews to wash in rivers together with Christians. Thus it is only in modern times that the word *miqveh* has come to designate an installation that is solely used for ritual purification of a person or an object. The functional development of this installation requires therefore clear distinctions between the modern term, the ancient term, and the installation that is being described. This study uses the term ‘stepped pool’ rather than ‘*miqveh*’ so as to avoid confusion regarding the changing meaning of the term. This generic term is not meant to qualify the functional purpose of the installation.

ARCHAEOLOGICAL EVIDENCE

Despite its modest dimensions and rather plain features, few other sites in the ancient Near East have received as much scholarly attention as Qumran. Entire books, chapters and numerous articles pretend to solve the archaeological mysteries of the site. After de Vaux (1953, p. 83-106 and 541-561; 1954, p. 206-236; 1956, p. 533-577 and 1973, p. 1-45) most of this work has been carried out by historians, some specializing in the literature of the Dead Sea Scrolls (Eshel 1996, p. 131; Puech 1998, p. 21-36; Regev 2000, p. 222-245; Schiffman 1994, p. 37-61; and Stegemann 1998, p. 34-55), others in unrelated areas (Davies 1988, p. 203-207, and Golb 1980, p. 1-24). Surprisingly few archaeologists have tackled the problems of this site.



Fig. 1 – Stepped pool (locus 48/49) during de Vaux’s excavation.

Qumran’s stepped and plastered pools

In an attempt to avoid premature chronological and typological attributions, the following description of the plastered installations and their interconnecting channeling system will focus on purely structural aspects. Though it is likely that at one point during the site’s occupation the entire pool complex was in simultaneous use, according to de Vaux’s interpretative reconstruction this occurred during period IIb (100-31 B.C.), it is almost impossible to reconstruct the different stages preceding the water system’s definitive form. For those installations where minor structural changes or additions alter the original aspect of a single pool, notification is made.

Aside from de Vaux’s general reports (1973, p. 1-45), it is Strobel’s (1972, p. 57-65) detailed descriptions of the water installations that are most valuable to us. Those descriptions are the result of his visits to the site during the summer of 1965, preceding the work of the Jordanian conservation team. The present author’s own measurements and field observations, as well as the plaster sampling (see geo-chemical plaster analysis by Aryeh Shimron), were carried out shortly before additional conservation work, mostly on the plaster, further obstructed the original state of the installations.

The water installations (pools, basins and channels) cover approximately 13,5 percent of the site’s built-up surface⁵. The volume of the pools varies between 2,000

and 204,000 liters (table 1 and fig. 24 to 28)⁶. On the general top plan (fig. 2) 26 plastered installations can be distinguished:

- one circular pool, locus 110 (5,10 m in diameter);
- three enormous rectangular pools (between ca. 12,5-17,5 m in length), two of them stepped (loci 56/58, 71) and one without steps (locus 91);
- four average-size rectangular pools (loci 48/49, 43, 117, 118 and 138, (between ca. 1,5-5,0 m in length), all of which are stepped;
- six small pools (between ca. 1,5-2,75 m in length) with steps (loci 50, 67, 68, 69, 83 and 85);
- six small pools (1,75-2,0 m in length) without steps (loci 55, 57, 70, 72, 119 and 119 bis);
- and finally four small basins (locus 34), one with steps, the other three without.

Loci Numbers	According to North	According to Laperrousaz	According to Galor
34	1.35	–	0.6
43	–	–	4.0
48/49	38.64	56.0	40.0
50	5.4	5.0	6.0
54	–	–	2.0
55	–	10.5	6.0
56/58	99.0/90.0	53.0/140.0	30.0/85.0
57	–	11.5	5.0
67	1.8	4.0	2.0
68	2.64	9.0	2.5
69	2.4	10.0	7.0
70	1.6	6.0	3.0
71	225.0	332.5	300.0
72	–	8.5	2.0
83	3.52	5.5	5.0
85	8.4	3.0	5.0
91	162.0	260.0	250.0
110	54.5	–	110.0
117	46.25	50.0	60.0
118	38.75	40.0	40.0
119	–	–	1.0
119bis	0.28	–	1.0
138	2.21	125.0	25.0

Table 1 – Volume of plastered installations in m³.

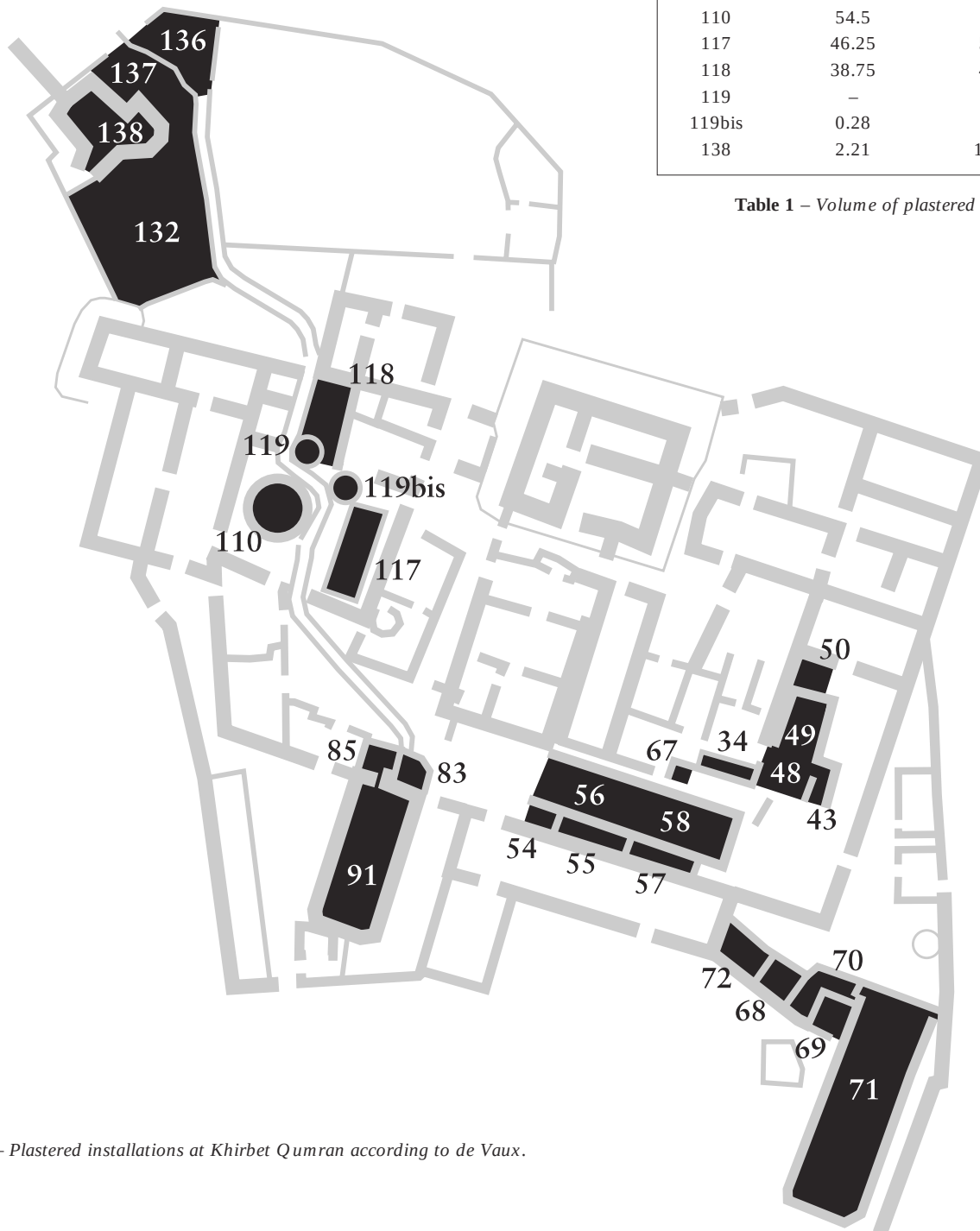


Fig. 2 – Plastered installations at Khirbet Qumran according to de Vaux.

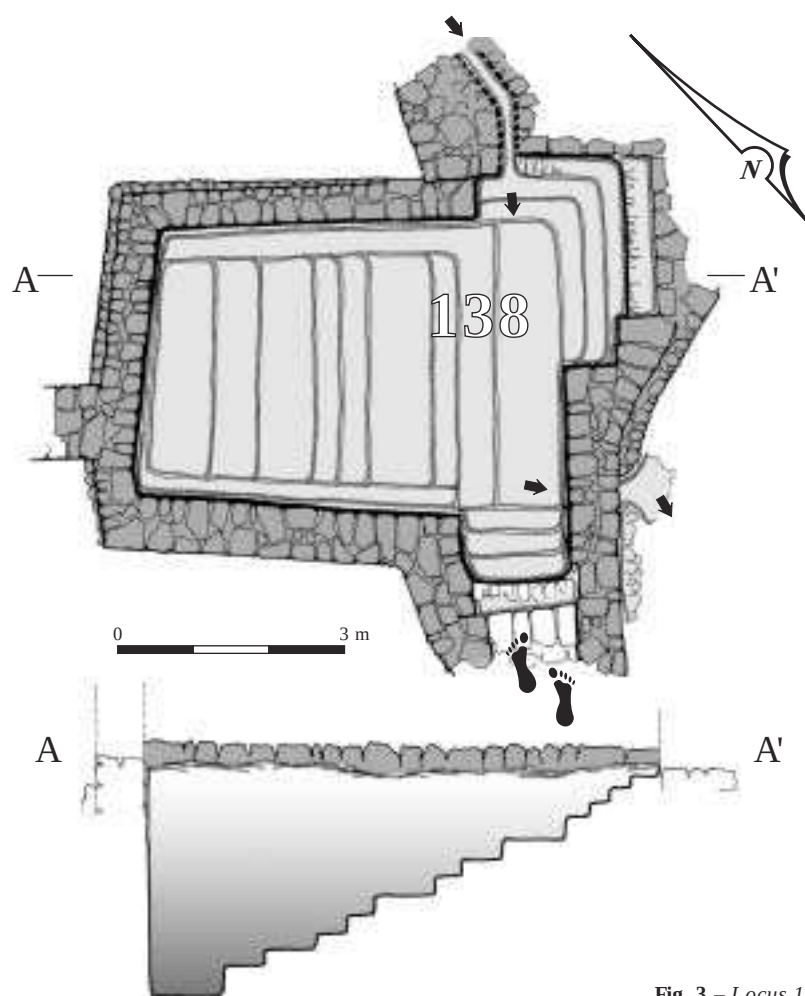


Fig. 3 – Locus 138: plan and section.

In conjunction with the water installations at the site, an aqueduct was built to divert the water from the wadi to the building complex. This aqueduct, partially dug and partially built into the marl terrace, connected a catch basin located at the head of the wadi and the settlement located some 700 meters away⁷. At the site the aqueduct continues as a channel covered with a similar plaster used to render the pools and basins impermeable. Some of the stone slabs that were used to cover the channel can still be seen scattered throughout the site.

As the aqueduct reaches the northwestern corner of the building complex, it first connects with locus 136 which according to de Vaux, served as a catch basin. Water would then flow from 136 through 137 into 138. According to de Vaux, locus 132 served as decantation basin (1959, p. 9). Though de Vaux's definition of each locus, loci 132, 136 and 137, in isolation would have been possible, it would not have functioned as a unit. A detailed study by Humbert and Galor, of the water system with its functional and chronological aspects will appear in volume III of the final report series.

A stepped pool, locus 138 (fig. 3), occupies the northwestern extremity of this large shallow basin (preserved to the height of 0,2-0,5 m). The pool can be accessed separately via four steps encompassing the eastern corner of the installation and another four steps in the southwest. These two staircases connect to a wider staircase (ca. 2,7 m) consisting of eight steps leading down towards the basin proper (width 2,2 x length 0,9). About 1,8 m above the bottom level of the pool, two horizontal shelves (ca.

0,4 m) flank the wide staircase. The water came through a circular channel debouching above the westernmost end of the top wide step. One would approach the pool through a door located to the east of the northern end of locus 138. An additional approach was possible from the eastern courtyard, locus 135.

The aqueduct channel winds around the eastern side of the stepped pool, locus 138, and then again around the eastern extremity of locus 130 to finally reach the northern end of a circular pool, locus 110 (fig. 4). As the channel circumscribes the eastern half of the pool's circular surface (ca. 5,10 m diameter) it opens from the south, probably supplementing an earlier channel reaching the installation from the north. In order to fill up the pool from the south, the walls had to be built up by at least 0,75 m. To the northeast and southeast of the circular pool (locus 110) another two stepped pools, loci 117 (fig. 5) and 118 (fig. 6), lie on an approximate north-south axis.

Locus 119, a small (width 1,8 x length 1,8 m) square pool, is located in between loci 118 and 110. Prior to the construction of the aqueduct, water could flow below locus 119 to locus 110 directly through a channel. During a later stage, locus 119 bis was built as a small basin adjacent to locus 118.

In locus 118, eight preserved steps lead from south to north down to the basin (width 2,4 x length 1,9 m). South of the preserved steps de Vaux reconstructed three additional steps. The incomplete state of the staircases' uppermost section, combined with the fact that loci 118 and 119 were enclosed by walls from all sides, make it dif-

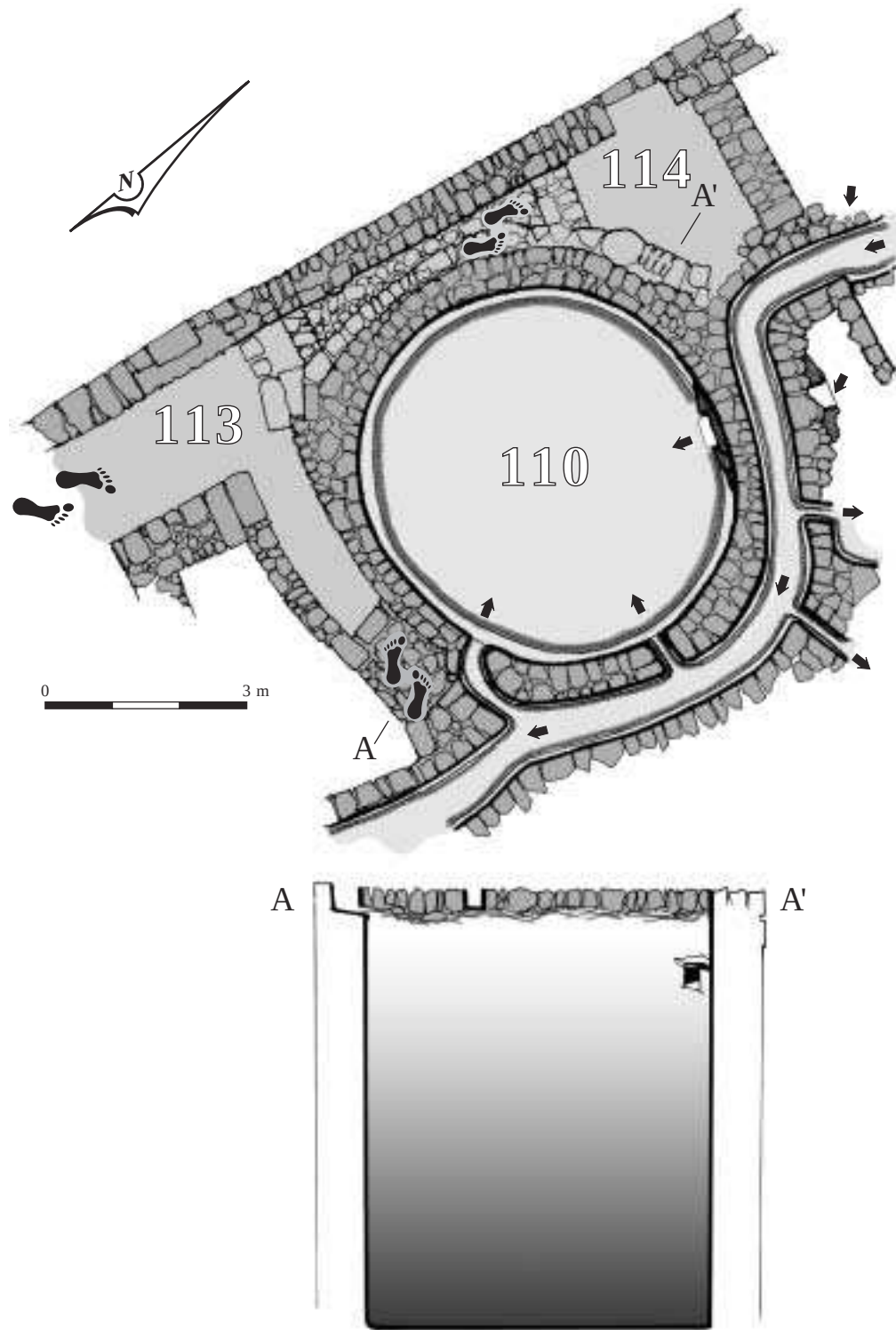


Fig. 4 – Locus 110: plan and section.

difficult to reconstruct the pool's original access. The water would reach the pool from the channel near the southeast corner, and flow down the steps gradually filling up the basin and ultimately the entire pool.

Locus 117, similar in plan and size to locus 118, can be entered from the top step in the north and descends towards the south via a series of 13 steps. The five uppermost steps are subdivided into three lanes of approximately equal width by two low stepped partitions. An auxiliary step (width 0,45 x length 0,35 m) simplifies access into the bottom basin (width 2,4 x length 1,8 m).

The four evident water channels (three from the west and one from the east) make it difficult to determine how the pool was supplied during its different periods. One of the water channels coming from the west was clearly there before the aqueduct.

To the south of the circular pool (locus 110) the aqueduct continues in a southeastern direction to connect with the two small stepped installations, loci 83 and 85 (fig. 7), and the large stepless installation, locus 91 (fig. 7), located immediately to the south and corresponding in its width to the combined width of the two stepped pools.

Locus 85 can be accessed via four steps leading from north to south towards the bottom basin (square 1,7 x 1,7 m), which opens directly on the 4 meters deep pool 91. Water entered the pool by an aperture connected with the basin 83, through the east side of the pool wall. This supplementary channel continues towards loci 77 and 86

of the entire pool's width (length 2,5 x width 1,6 m). An additional channel is located at the top of the eastern wall.

The main channel continues in an eastern direction from locus 83 towards the large, rectangular installation of loci 56/58 (fig. 8). This installation stretches parallel to the so-called refectory, locus 77, separated by an additional

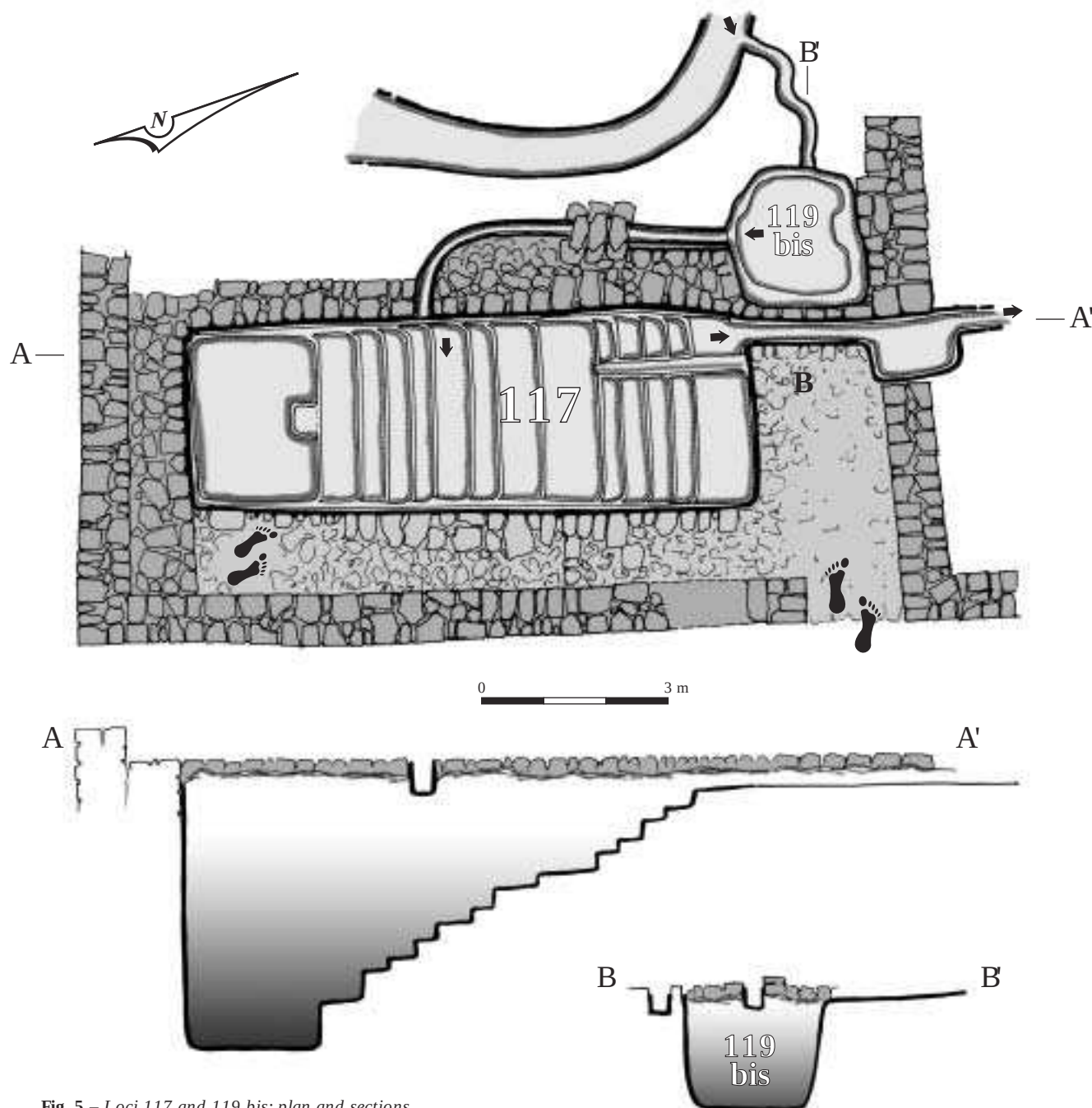


Fig. 5 – Loci 117 and 119 bis: plan and sections.

stretching partially over a fill occupying the northern part of the large rectangular pool, locus 91 (length 12,6 x width 4,6 m), therefore belonging to a later stage of the system's development. It cannot be excluded that the original access to the stepless pool, locus 91, was via the steps leading down into locus 85. However, only additional excavations removing the fill supporting the supplementary channel in the northern part of locus 91 would be able to clarify this point. Locus 83, accessed via three steps, was directly connected to the main channel situated to its north. The steps of this pool only occupy about half

series of narrow pools, loci 54, 55 and 57. The bottom of loci 56/58 in its original stage (length 17,0 x width 3,3 m) could be reached by descending a total of 11 steps from west to east. The five uppermost steps seem to have two low stepped partitions placed parallel and south of the mid-point. During a subsequent stage the installation was subdivided into two, and ultimately three, separate parts. A first wall (height 1,7 x width 0,8 m) crossing the original installation on a north-south axis created a stepped pool to its west and one (later two) stepless pool(s) to its east. The bottom basin of locus 56 (length 1,7 x width

3,5 m) could be reached by using an auxiliary step, built after the construction of the original staircase (spanning the entire width of the installation) and possibly in conjunction with the construction of the dividing wall. The additional secondary wall inside the stepless pool, locus 58, built to the east and parallel to the main dividing wall, is only preserved in its three bottom-most courses. De Vaux cleared most of its northern part (Humbert and Chambon 1996, p.311).

Shortly before reaching the westernmost step of locus 56, the main channel forked into three different directions. One branch in the south leads towards locus 77, approximately 0,25 m above floor level, and located some 1,2 m to the east of the door opening into this space. A second branch led the water flow directly into the stepped pool,

water reached locus 54 through a small channel located close to the southwestern corner of locus 56.

The square installation (width 1,4 x length 1,6 m), locus 67, could be entered from its southwestern corner via a small auxiliary step. It is from the same corner that the water would have flowed into the installation. Locus 67 might have supplied water to the small basins of locus 34 prior to their functional transformation (see description of locus 34).

The main channel winds around the northeastern corner of locus 58, continuing in a southward direction connecting consecutively to four small pools, loci 72, 68, 70 and 69 and one large pool, locus 71. From the northeastern corner of locus 58 a secondary channel branches off in a northward direction supplying loci 48/49 (fig. 9) from the south.

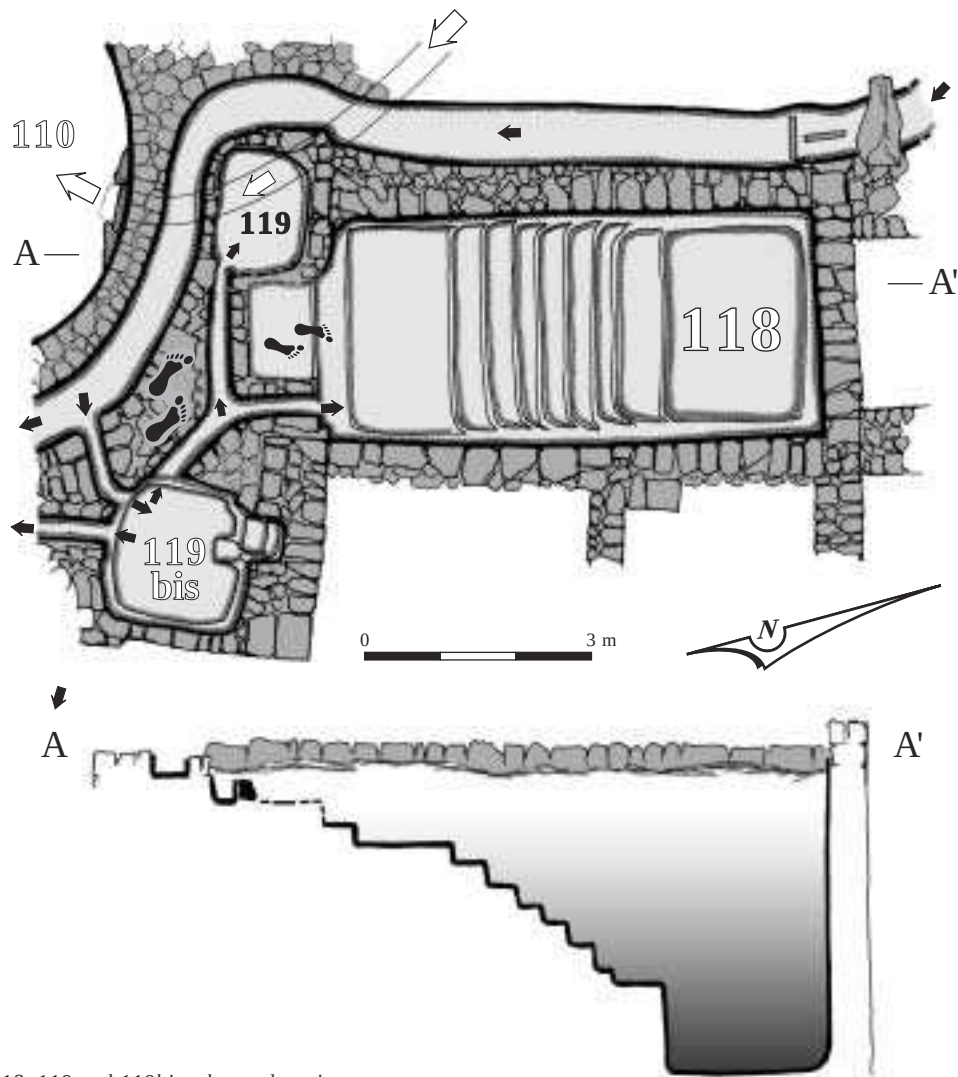


Fig. 6 – Loci 118, 119 and 119bis: plan and section.

locus 56. Finally a third branch to the north stretches along the entire length of loci 56/58 and filled up loci 67 and 58.

The elongated space (depth approximately 1,8 m) in between locus 77 and loci 56/58 was subdivided into four shorter spaces, loci 54, 55 and 57⁸. It is impossible to reconstruct how these installations would have been supplied with water. However, the fact that there are remnants of plaster, as well as their proximity to loci 56/58, suggest that they might have contained water at some point. De Vaux's reconstruction suggests that the

The channel widens into the southwestern corner of loci 48/49, flowing down the 14 steps from south to north leading into the basin proper (length 1,1 x width 2,7 m). A slightly detached auxiliary step facilitates descent into the pool. According to de Vaux (1959, p. 20), an earthquake was the cause of the split visible in the center axes of the pool. The result is the difference in height between the eastern and western halves of the pool (length 7,2 x width 2,7 m)⁹. The eastern half is some 40 to 50 cm lower than the western half. The five uppermost steps are subdivi-

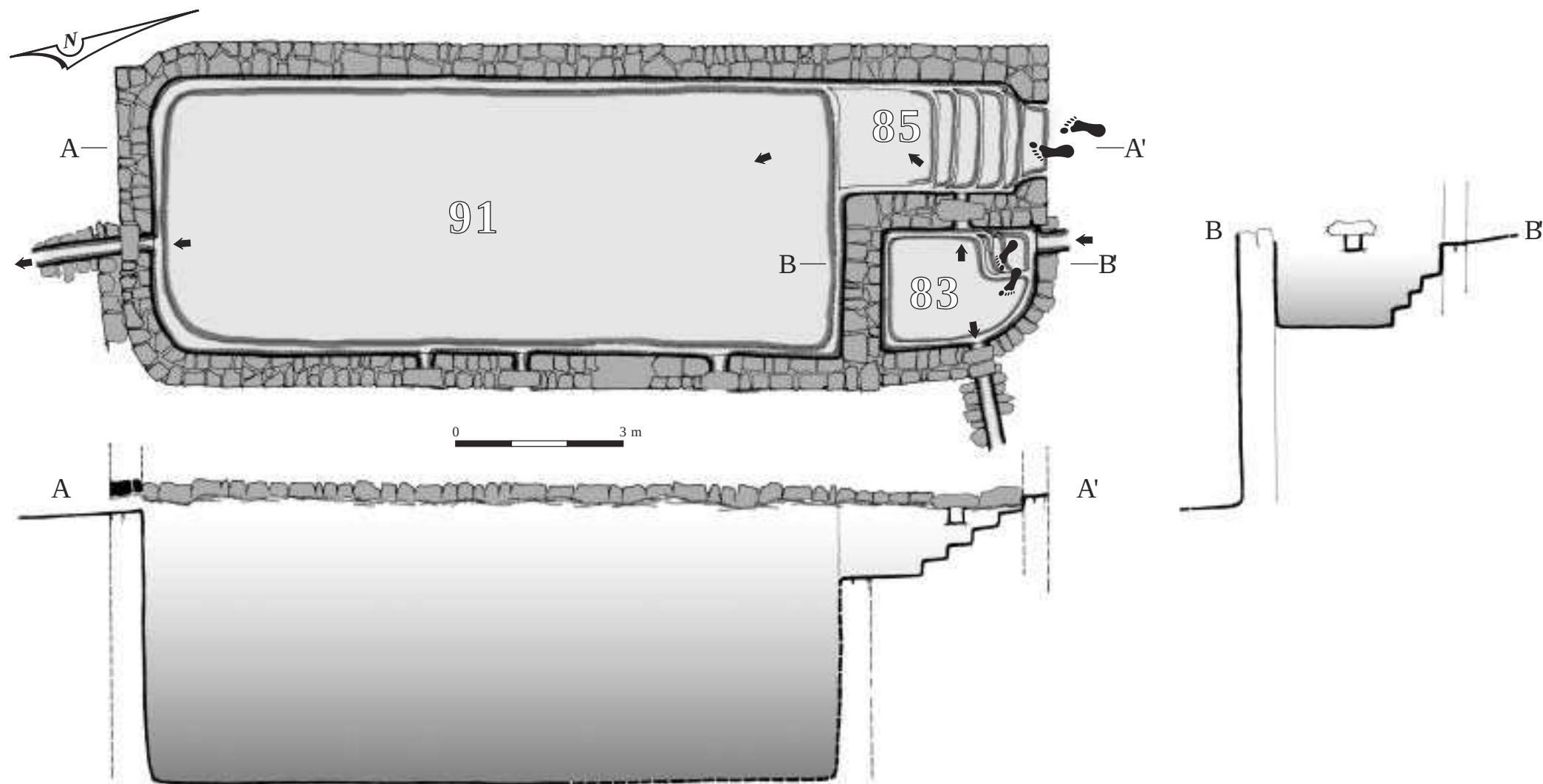


Fig. 7 – Loci 83, 85 and 91: plan and sections.

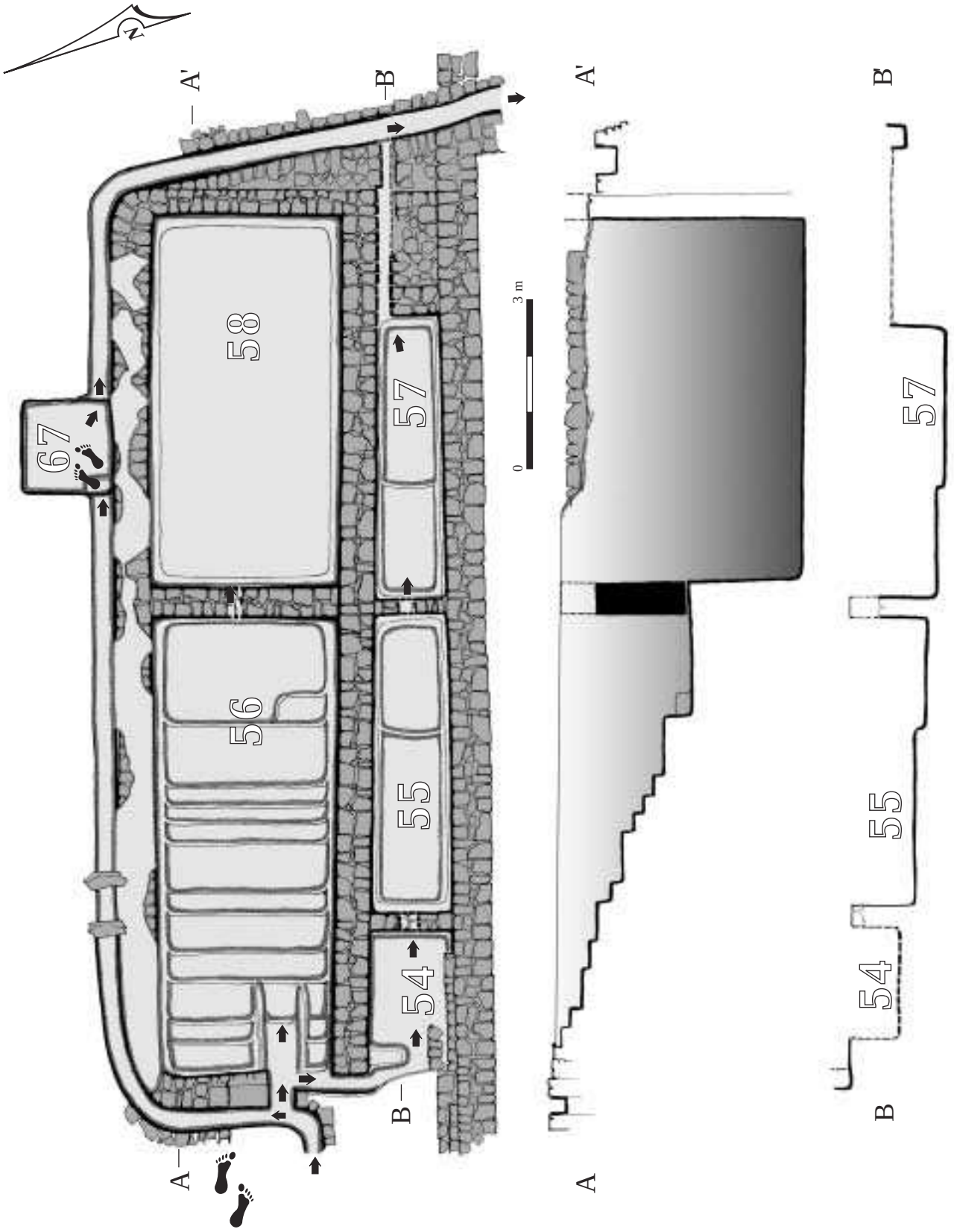


Fig. 8 – Loci 56/58: plan and sections.

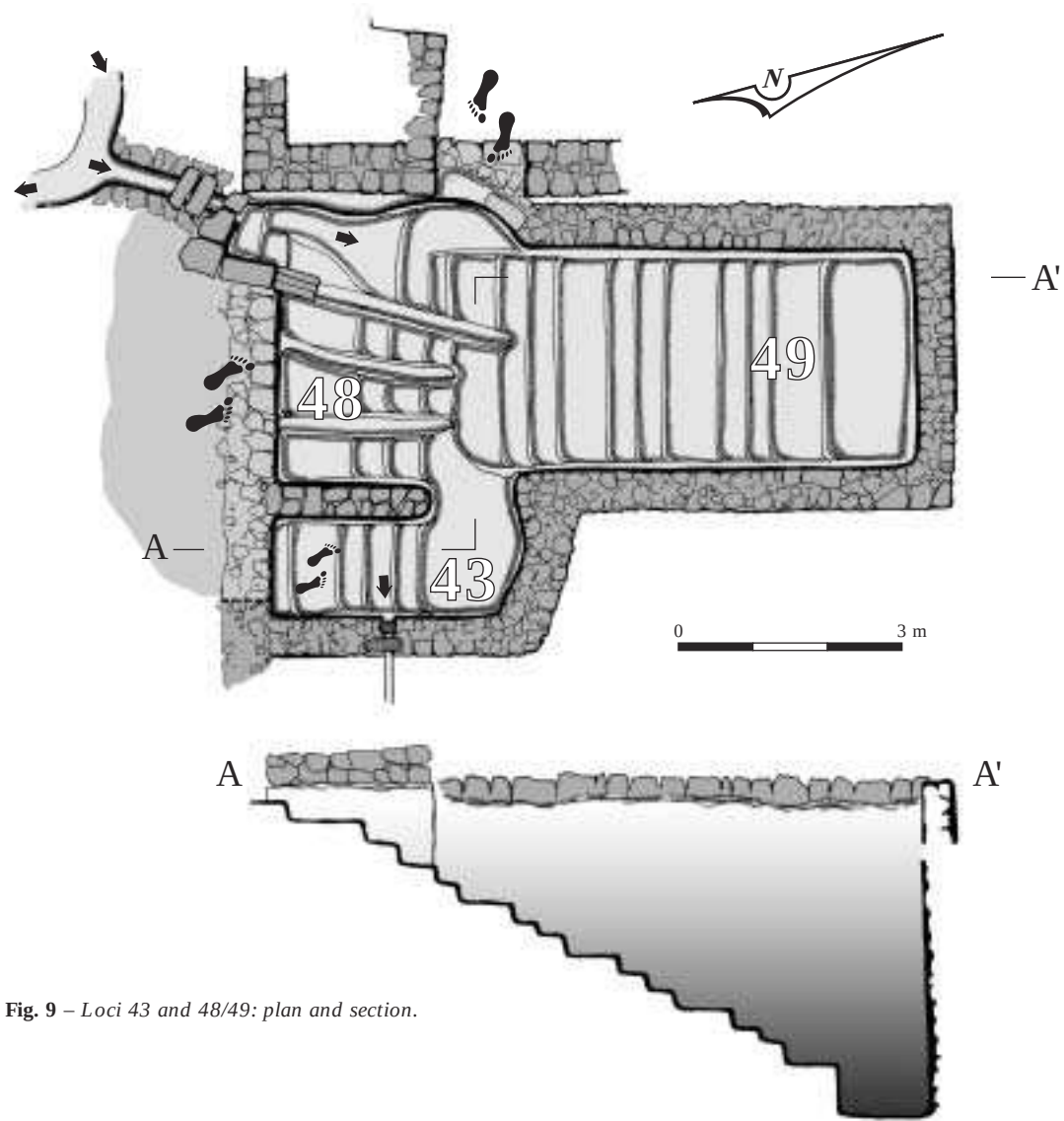


Fig. 9 – Loci 43 and 48/49: plan and section.

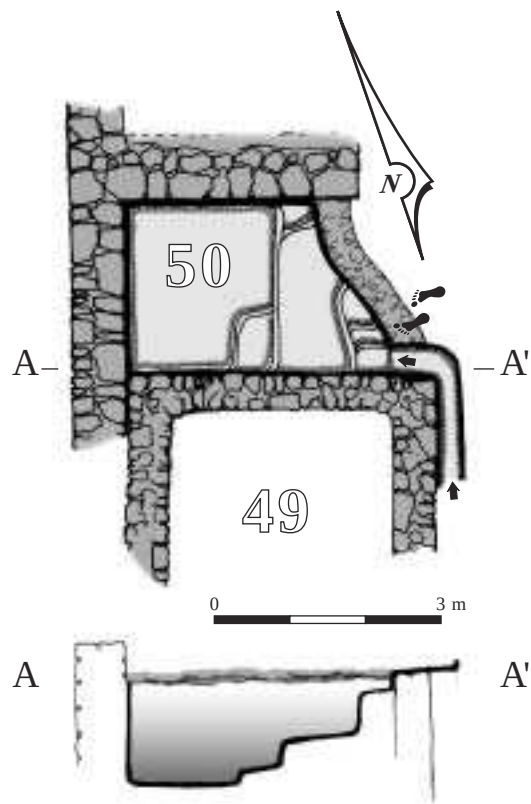


Fig. 10 – Locus 50: plan and section.

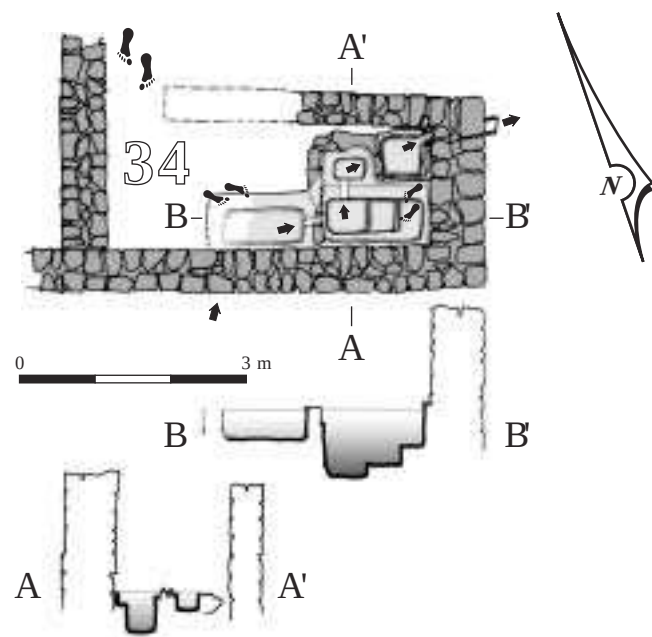


Fig. 11 – Locus 34: plan and sections.

vided by three low stepped partitions. An additional entrance (locus 43) separated by a wall from the main steps provides further access to the pool. Five steps descending from south to north give onto a landing, which joins the main steps from the west. This landing, or bench, is mirrored on the western side of the pool by another horizontal surface of the same height. From the eastern landing/bench an auxiliary step (width 0,7 x length 0,5 m) facilitates descent into the basin proper (width 2,4 x length 1,3 m).

Locus 50 (fig. 10) consists of a square-shaped pool (western side 2,2 m, northern and southern sides 2,1 m) with a rounded stepped addition on its eastern side. This addition is made up of three narrow steps (width 0,16-0,38 m) descending from south towards north. De Vaux

Whether used at this later stage as a dyeing installation, as suggested by de Vaux, or for some other domestic or industrial function is unclear. However, their primary use as bathtubs can be substantiated by comparing the two southernmost basins in their original state to similar installations known from other contemporary sites. Bathtubs of similar shape and size were found in the Late Hellenistic/Early Roman dwellings in the Jewish Quarter of Jerusalem (Avigad 1980, p. 104, 154 and 160), as well as in the Hasmonean palaces of Cyprus (Netzer 2001, p. 73) and Jericho (Netzer 2001b, p. 32 and 33)¹¹.

At the same place where the secondary channel branches off to the north towards locus 48/49, the main channel continues to the south reaching the southeastern corner of locus 77. From here the channel takes a southeastern

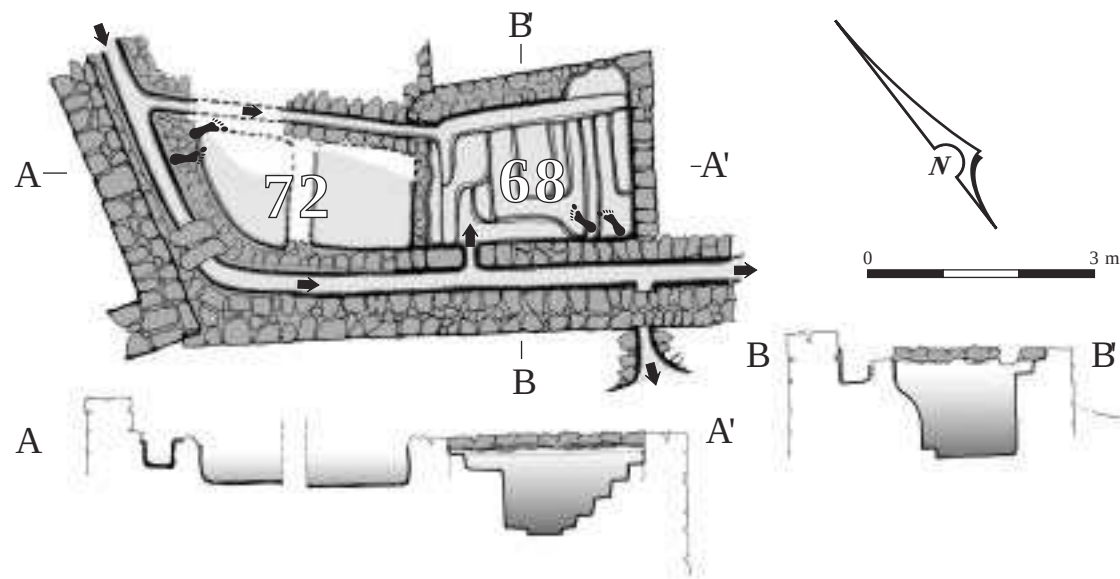


Fig. 12 – Loci 68 and 72: plan and sections.

(1959, p. 20) and Strobel (1975, p. 63) describe the earthquake dislocation within locus 50, which is perfectly aligned with the split in loci 48/49.

The two tub-like structures and small basins in locus 34 (fig. 11), only briefly mentioned by de Vaux (1959, p. 7 and 16), are consistently ignored by all those treating the subject of the plastered installations¹⁰. Upon discovery three steps descended into the eastern tub (length 1,4 x width 0,5 m) from its eastern short end. The fragmentary state of the western pool, probably due to a reuse of the space for a different function, does not allow us to accurately reconstruct its original dimensions (ca. length 1,3 x width 4,2 m). The overflow of both installations was diverted towards the northeast, initially flowing through a rectangular stone channel that eventually connects to a clay pipe. As indicated by the partially destroyed western tub, and the functional change of the eastern tub, the initial use of the installations belonged to an early stage of the site's settlement. This early date of the tubs' original use as water installations can further be substantiated by the fact that the overflow channel passed in place of loci 48/49, and was therefore clearly built prior to the latter installation. It is likely that the functional transformation of the tubs and basins occurred after the construction of loci 48/49.

direction supplying a continuous row of small installations, loci 72 and 68 (fig. 12), 70 and 69 (fig. 13).

The westernmost stepless installation, locus 72 (length 2,7 x width 2,0 m), adjoins a stepped pool, locus 68 (length 3,0 x width 2,0 m). Once locus 68 was built, locus 72 went probably out of use. A series of irregularly displaced steps starting off from east and west lead down into a small bottom basin (length 0,5 x width 1,2 m) of locus 68. A narrow shelf extends along the northern side. The water entered the pool from its southwestern corner. The direction of access to these pools is no longer evident.

An additional stepless pool, locus 70 (length 2,7 x width 1,2 m), is positioned in between the two stepped installations, loci 68 and 69. As in the case of locus 72, following the construction of locus 68 locus 70 also went out of use. Its northern extremity is bordered by a flat basin (length 3,7 x width 1,7 m) oriented on an east-west axis. The proximity to the potters' workshop situated to the south as well as the fork/diversion of the main channel leading to locus 75 should be mentioned, although a clear functional connection cannot be proven.

Immediately to the east of locus 70 lies locus 69 (length 2,72 x width 2,5 m), supplied with water by the channel penetrating the installation from its southwestern corner.

Although two flights of narrow steps (0,6 m) extend along the eastern side of the pool, one descending from north to south, the other from south to north, with a meeting point midway down the eastern wall, the installation could only be accessed from the south. Two auxiliary steps, protruding into the basin (width 2,5 x length 1,6 m), facilitate the final descent into the installation; the top auxiliary step is subdivided at the midpoint by a low partition.

Locus 71 (fig. 14) constitutes the southeastern extension of the building complex at Qumran. It is the largest

tered installations is particularly problematic because no traces remain of covering structures, which constitute a most essential part in water-containing installations.

Given the large number of well-preserved parallels for the average and large size pools, not only in Palestine, but all over the Mediterranean, the most plausible reconstruction for the installations at Qumran would include a covering structure. Contemporary storage and immersion pools as well as cisterns (with or without steps) were normally covered, either with vaulted ceilings, beams,

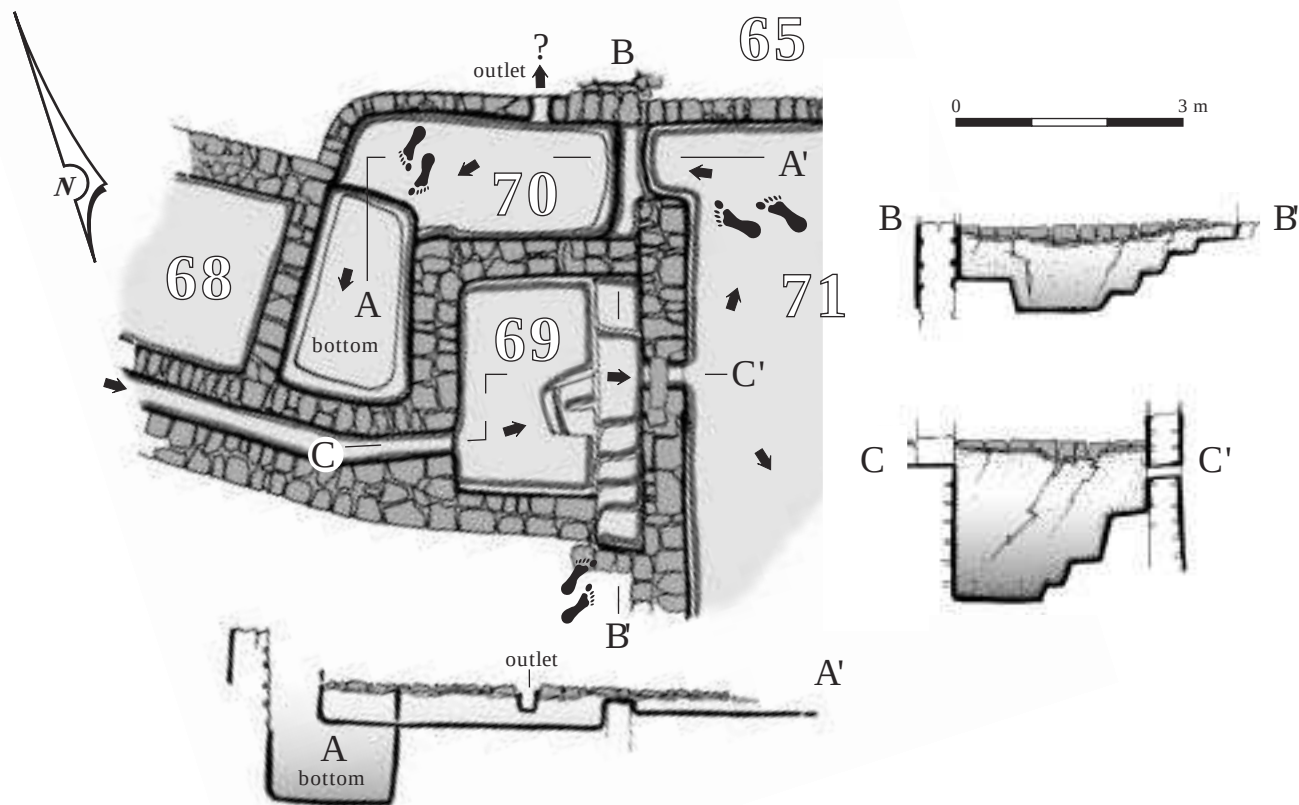


Fig. 13 – Loci 69 and 70: plans and sections.

stepped pool (length 17,4 x width 4,5 m) at the site with 20 steps extending over the entire pool width and leading from north to south into the basin (length 6,4 x width 5,3 m) proper. Water coming from locus 70 or locus 69 would first reach a flat basin (length 1,44 x width 5,3 m) to the north of the steps, and then flow down towards the bottom of the installation.

Interestingly, de Vaux and Strobel hardly ever agree upon how the water entered the different pools, and more importantly, how the water was evacuated¹². According to my own observations, those features that are described as evacuation or overflow channels do not seem to correspond in the least to the characteristics usually associated with this kind of structural element. In loci 71 and 91, what has been identified as an overflow channel is positioned at a higher elevation than the supply inflow.

Although meticulous recording and description of existing features and structures constitutes the archaeologist's most important task, his interpretative work usually also relies on elements that have vanished and need to be reconstructed. The partial preservation of Qumran's plas-

capstones or possibly perishable materials¹³. This was done not only to avoid evaporation, but also contamination of the water by dirt, dust, sand and algae formations. Other factors, such as security and the optimization of building and living space, cannot be neglected either. The fact that no traces of superstructures have survived at Qumran led de Vaux to the conclusion (1959, p. 9) that the cisterns had never been covered. Although he realized that the absence of cover would cause the contents to evaporate very quickly during the summer, he proposed no solution. Wood (1984, p. 57-58), fully aware of the high evaporation losses, particularly in the desert climate¹⁴, suggests that the pools were covered with wooden or skin covers, or possibly with mats made from the reeds growing at 'Ain Feshkha.

De Vaux was certainly right in not believing that the pools at Qumran were covered with a corbeled or vaulted structure. The use of well-dressed stones would have left clear traces within the fill removed during the excavation of the installations. Wood's proposal that the pools were covered with some sort of organic material is most con-

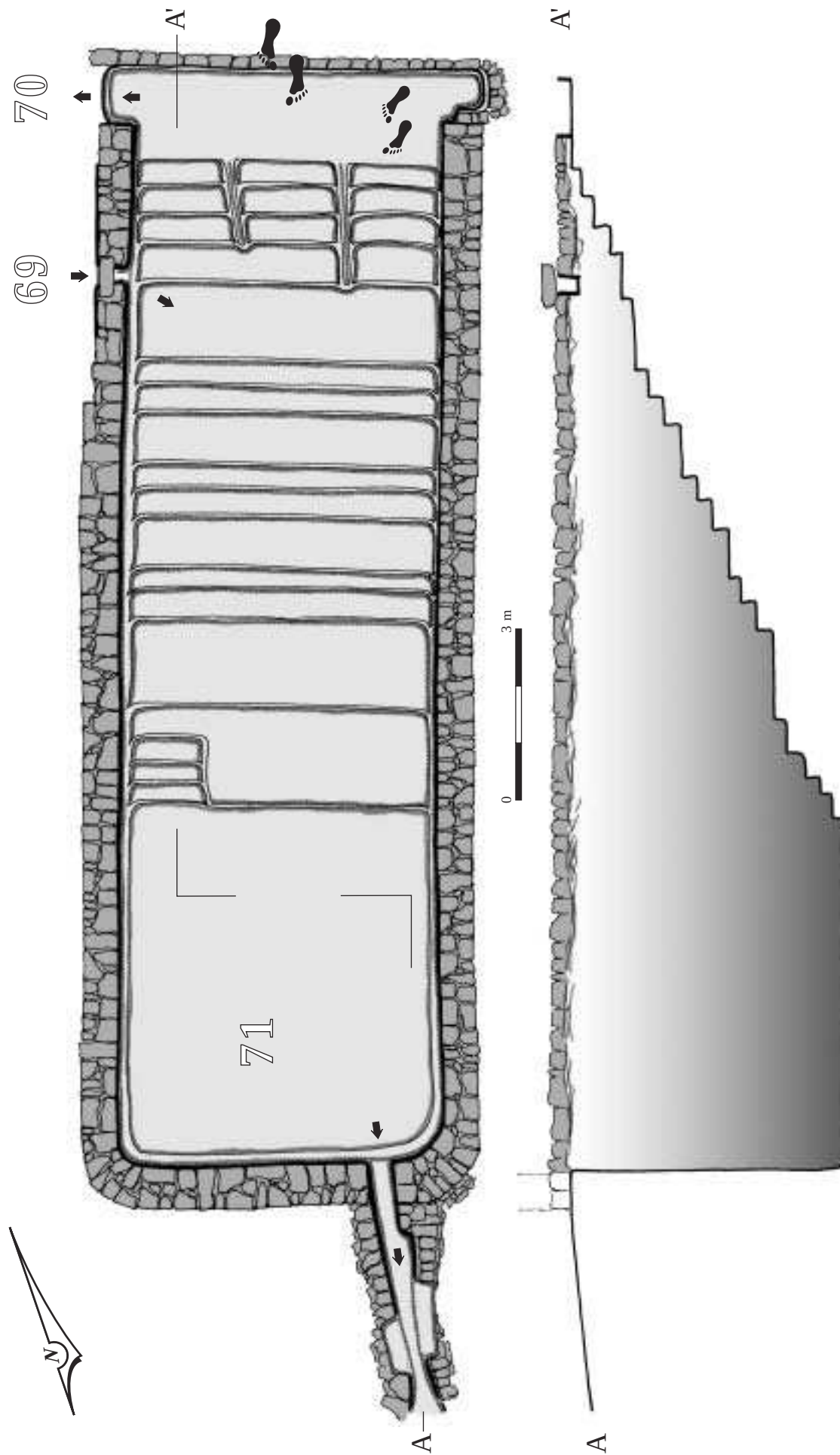


Fig. 14 – Locus 71: plan and section.

vincing. Fragments of palm tree trunks were found in loci 4 and 13 (Humbert and Chambon 1996, p. 293 and 297). It is likely that the wood of this tree fulfilled some structural function. With a maximum length of 9 meters, the trunks of date palms could have easily spanned the widest installation at the site¹⁵. The archaeological evidence¹⁶ of the use of palm-trees in ancient constructions is well attested all over the Near East (Postgate 1980, p. 99-101). Biblical and Talmudic sources (Taglicht 1917, p. 4907-416) attest to the multiple virtues and uses of the date palm within Palestine¹⁷.

For those pools that were not located within a roofed area (most likely loci 71, 91, 110, 117, 119 and 138) a single row of palm trunks would have covered the pool. Pools located within the roofed parts of the buildings (most likely loci 48/49 and 56/58) would be covered immediately at ground surface, and once more at ceiling or roof level. For regular roofs, that is the ceilings covering the rooms or supporting an additional level, the trunks would be covered with palm branches, reeds or rushes (Postgate 1980, p. 101-107). Those trunks that were placed immediately above the water installations would be exposed to a much higher degree of humidity. Therefore, the usual thin application of earth applied on top of the above-mentioned organic materials would be considerably thicker, and if rich in clay would have provided better insulation¹⁸.

Typological observations

The above description categorizes the plastered installations typologically according to size, shape and accessibility. De Vaux categorizes the pools mostly according to size and attributes different functions to each typological group (fig. 15). According to his assessment, all the large pools, both with and without steps, were used as cisterns (loci 48, 49, 50, 56, 58, 66, 71, 91, 110, 117 and 118); the smaller and shallower pools, with and without steps, were interpreted as decantation basins (loci 69, 83, 119bis, 132 and 137); and without offering an explanation, two of the stepped pools (loci 68 and 138) were described as baths (1973, p. 132). Petit (1981, p. 97) adopted de Vaux's typological suggestions, also viewing loci 68 and 138 as the only installations serving the purpose of ritual immersion.

It was North (1962, p. 107) who first noticed the importance of the reservoirs' steps and the proportionate volume those occupied. In his idealized diagram (1962, p. 114), he was able to show that up to 40 percent of the storage space was lost on account of the stairs.

In Strobel's opinion (1972, p. 62), it is not only the architectural features such as low partitions and "generously built" staircases that characterize the Qumran pools as ritual pools, but mostly the fact that they were interconnected with the same channeling system. His claim is that each of the seven main groups of the plastered installations adheres to the strict rules of ritual immersion as formulated in the Mishnah¹⁹. Strobel tries to show with his calculations that, within each group the volume capacity of one pool always corresponded precisely to the minimal requirement of 40 *seahs*, translated by him as 525 liters (1972, p. 70-76). The size of the pool in which the actual immersion took place could, however, range from relatively small (such as locus 69) to very large (such as loci 117 and 118). The closest typological parallel to the system used at Qumran, according to him (1972, p. 70-72), are the Masada installations.

In the 1980s Wood (1984, p. 45-60) suggested differentiating between the stepless pools at Qumran used for storage of drinking water and the stepped pools at the site used for ritual immersion. He supports his theory with the fact that both the volume of water used for drinking, and the water used for ritual purification, grew proportionately from period Ia to period Ib. In other words, Wood not only assumes that the consumption of drinking water per individual can be calculated precisely but also the use of water for ritual purposes²⁰.

In his detailed study of ritual baths during the Second Temple period, Reich (1991, p. 47-61) comes to very similar conclusions regarding the typological divisions of plastered installations. In his general typology of ritual baths, most plastered pools that can be easily accessed via steps and that can hold at least 40 *seahs*²¹ of water can be defined as *miqva'ot*. Accordingly, the pools with no steps at Qumran (loci 58, 91 and 110) were interpreted by him as cisterns used for water storage, whereas all other pools with steps were used as *miqva'ot* for ritual immersion. The rather unusual distribution of the steps of locus 69 led him to assume that its function was slightly different. Supposedly, it was used for ritual purifications of furniture and other large objects (Reich 1999, p. 127).

Regev's typological classification not only has direct implications for the ritual function of the plastered installations but also for their users (1996, p. 4-12). In his opinion, those pools that are equipped with a double entrance, or a staircase divided by a low partition wall, were used by priests²². Accordingly, in the case of Qumran, this variety of pools (loci 48/49, 56/58 and 71) was used by the priestly members of the sect. For the usual single partition, he describes the functions of the two lanes, one being used for descent and one for ascent. Although he does not comment on the functions of the triple and quadruple lanes at Qumran, he could have adopted the same theory to relate the quadruple lanes to the four different sectarian ranks mentioned by Josephus²³. From a practical point of view, the interpretation of using the low partitions as a symbolic space divider does not make any sense. In the cases of single and multiple partitions, at Qumran and at other sites, the lanes created by the partitions are frequently as narrow as 0,15 meters.

Cansdale and Hidirolou reject the typological resemblance of the Qumran pools with other stepped installations identified as ritual immersion pools. Cansdale (1997, p. 128) mentions the existence of steps in the "ma'agura" cisterns in the Negev as well as in the Nabatean cisterns at Shivta, Nessana and Mamshit. Typologically speaking, the difference between the steps of the Qumran pools and the steps in these latter cisterns is rather striking. Hidirolou's (2000, p. 30-35) reference group for comparison with contemporary hydraulic installations consists of the Hasmonean palace-fortresses of Alexandrion, Dok, Cypros, Masada, Machaerus and Herodium. The installations at these latter sites, and their steps in particular, do not resemble the installations at Qumran.

The typological difference between the cisterns at these latter sites and the installations at Qumran and, more importantly, of their steps is not being considered in the least. The steps in the above-mentioned cisterns are constructed on one, or sometimes at two sides of the installations. The necessity of building steps was mostly connected to the desert climate, as they were meant to facil-

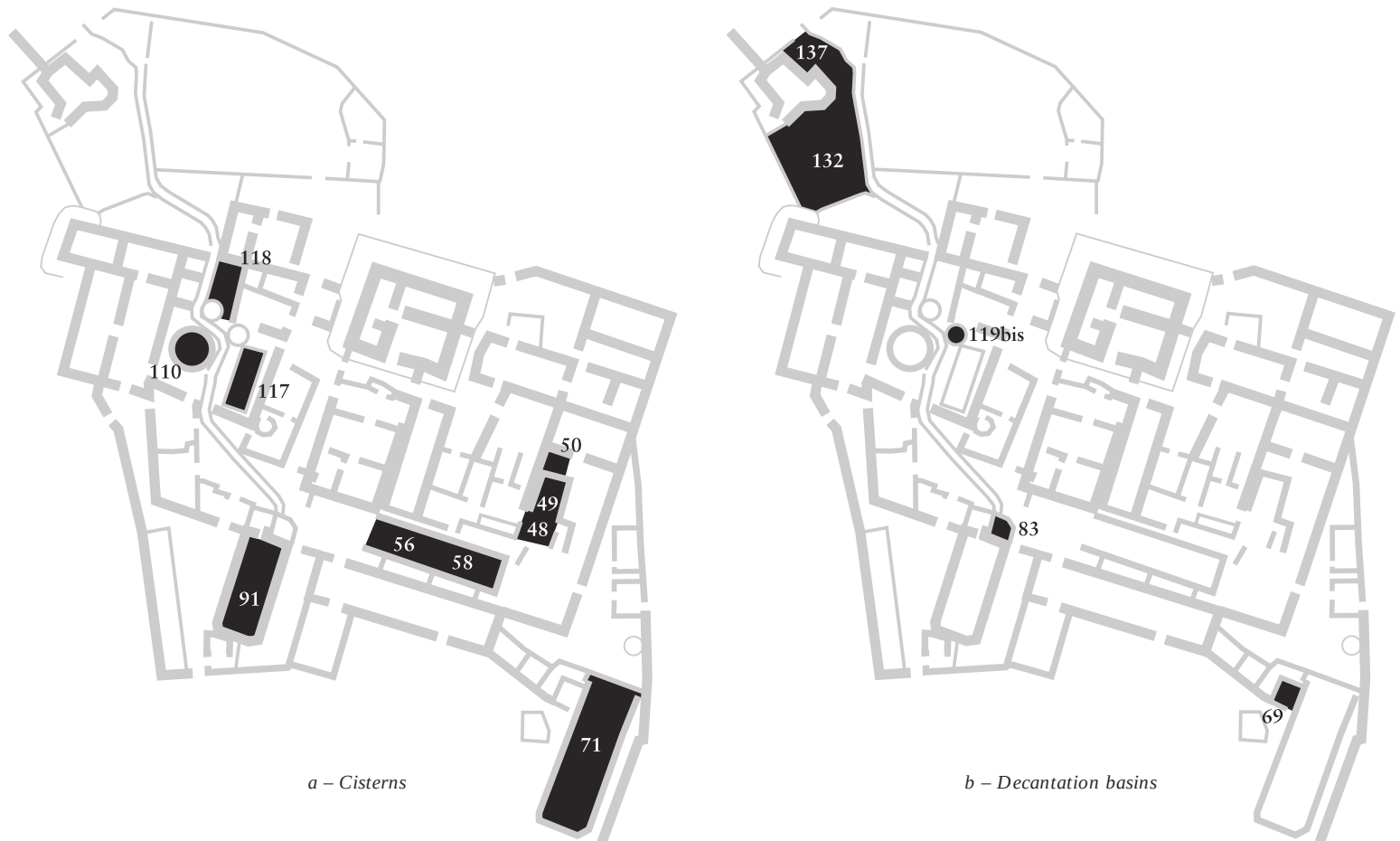


Fig. 15 – De Vaux's typological classification.

itate the drawing of water when the cistern emptied during the dry season (Evenari, Shanan, and Tadmor 1971, p. 159). Wood (1984, p. 47-49) and Reich (1990, p. 62-81) have both explained the main advantage of building a narrow staircase along one side of a cistern rather than across the entire width as is the case at Qumran. Narrow steps serve the purpose of drawing water just as well as wide steps. Furthermore, it is easier to build them, and they only take up about one-tenth of the volume of full-width steps.

Without a single exception, all the above-mentioned typological statements have direct implications for the functional interpretation of the Qumran installations. The diversity of the resulting interpretations is indicative of the flexible and therefore problematic nature of this methodology. It is relatively easy to point out examples and stress certain parallels and eliminate major differences that support one or the other theory. At this stage it would be more appropriate, however, to make typological statements without jumping to functional interpretations.

Beyond the typological observations regarding size, shape and accessibility of the installations, the pool walls themselves allow us to differentiate between different groups of construction types. Since the walls were originally entirely covered with several layers of plaster, the boulders, stones and mortars were not meant to be seen. The present condition of the plaster obviously barely resembles that of the 1950s and even less the original state. A comparison of pictures of the pools taken shortly after the excavations carried out by de Vaux with

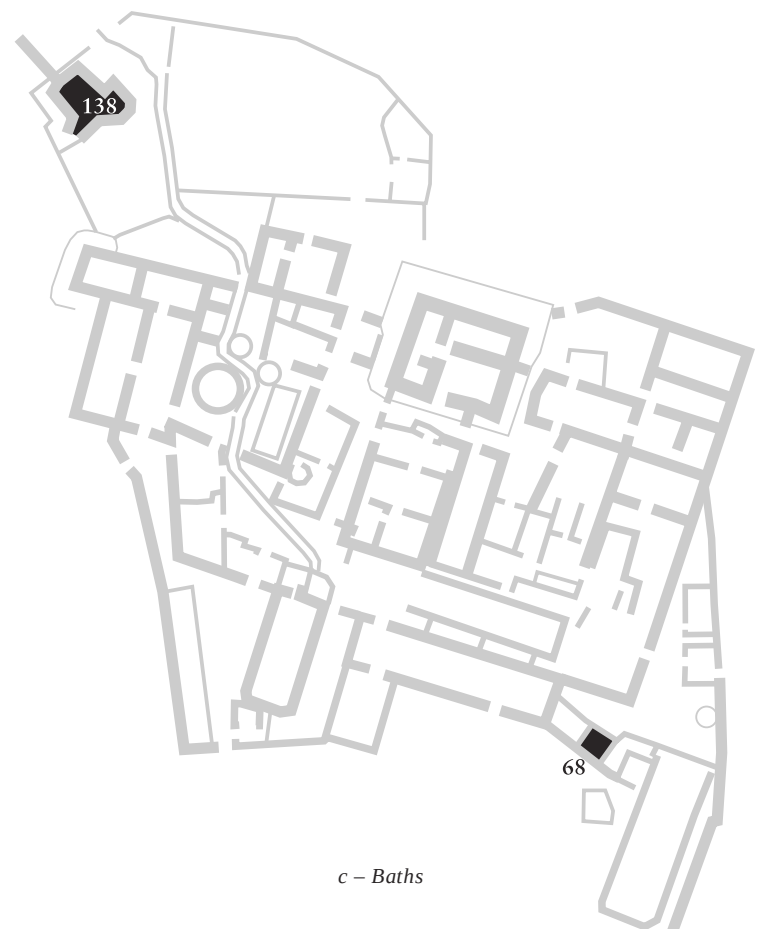




Fig. 16 – Galor's typological classification.



pictures of the pools today reveals significant differences in the quality, and especially the quantity of the plaster. In the 1950s, about 80 percent of the pool wall surfaces were still covered with plaster, whereas today about 70 percent of the plaster has disintegrated²⁴. This deterioration allows us to examine the parallels and differences between the various building types. All walls are made of the local limestone²⁵. The boulders and stones of the pool walls holding up the crumbly marl were placed in three different ways (figs. 16 and 20). 1) The walls of loci 48/49, 68, 83, 85, 91 and 138 are made of small boulders and fieldstones, placed randomly and held together with a coarse mortar (fig. 17). 2) The walls of loci 56/58, 69, 117 and 118 are built of irregular courses of small boulders filled in with smaller stones. These are also held together with a coarse mortar (fig. 18). 3) Locus 71 is the only pool where medium-sized boulders are placed in neat courses (figs. 19 and 21)²⁶. The spaces in between the boulders are filled in with smaller stones and a similar coarse mortar as the one observed in all the other pools. Although it is tempting to attribute the different building styles and their spatial distribution to sequential building stages, only a more refined contextual and stratigraphic report will be able to determine the origin of this typological variation²⁷. Rather than attribute the different building styles to a building sequence, one could attribute the differences to the non-homogenous character of the marl terrace (figs. 24 and 25).



Fig. 17 – Type 1 (locus 48)

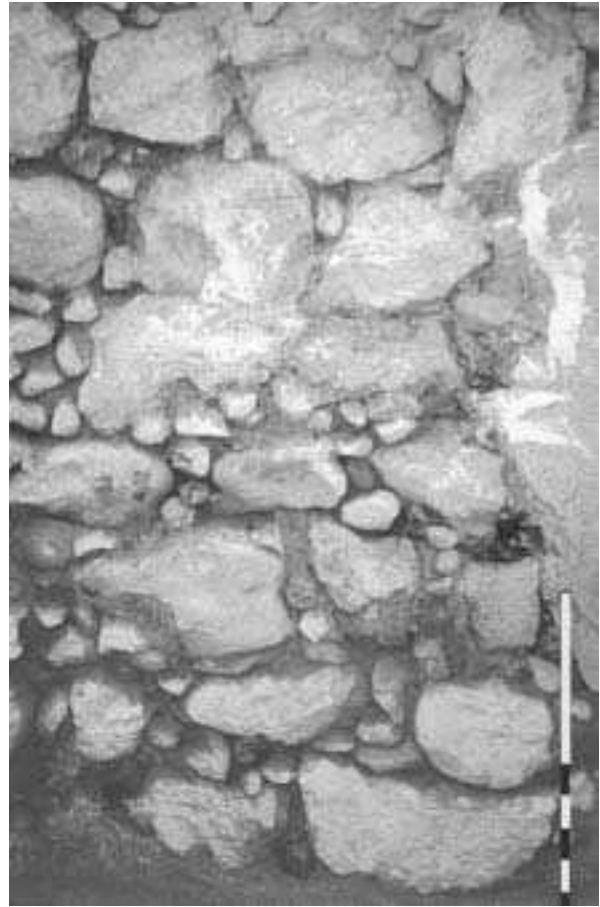


Fig. 18 – Type 2 (locus 118)

Chronological observations

De Vaux (1973, p. 1-45) divided the occupation of the site into five principal phases, starting with the Israelite occupation in the eighth century B.C. and closing with the end of the Second Jewish Revolt in 135 A.D. A detailed description of the water installations is carefully integrated into his account of the distinct periods.

According to de Vaux (1959, p. 2-3), the large round cistern (locus 110) was built during the Israelite occupation of the site (eighth and seventh centuries B.C.), and filled during the winter season by the surface runoff from the esplanade to the north of the settlement (fig. 22a). Two new stepped rectangular pools (loci 117 and 118) were added during Period Ia (dated by de Vaux from ca. mid second century B.C. to the time of John Hyrcanus, ca. 135-104 B.C.). During this stage (fig. 22b), a newly built channel brought water from the esplanade toward a decantation basin (loci 132 and 136), which in turn fed the refurbished round cistern and the two additional stepped pools (1959, p. 4). In the succeeding Period Ib (dated by de Vaux from the reign of John Hyrcanus to the earthquake of 31 B.C.), the buildings and water system reached their definitive form (fig. 22c). The aqueduct was built and the site itself received four additional stepped pools (loci 56/58, 48/49, 71 and 138), another stepless pool (locus 91) and a large number of smaller pools (loci 50, 67, 68, 69, 83 and 85). Finally, the two tubs in locus 34 also date to this period (1959, p. 5-7).

The earthquake occurring at the end of Period Ib left clear traces in two pools (loci 48/49 and 50), causing a dis-



Fig. 19 – Type 3 (locus 71)

location of several centimeters between the eastern and western sides of the installations (1959, p. 20). During the following Period II (ca. 1 B.C. to 68 A.D.), those two pools were abandoned (fig. 22d) and a wall was built across locus 56/58 (1959, p. 27). During Period III (68-73 A.D.), only one large pool (locus 71) remained in use (1959, p. 43). Being somewhat isolated from the main buildings, this installation has suffered the least damage (fig. 22e).

The two points in de Vaux's chronology that have received the most substantial criticism concern the date of the establishment of the Hellenistic settlement and the period following the earthquake²⁸. Since these chronological observations have no impact on the chronological analysis of the pools, they are not taken into account in this study.

Surprisingly, no one has questioned the chronological development of the pools as suggested by de Vaux. Even those scholars who have dealt in detail with the Qumran pools (North 1962, p. 110-116, Strobel 1972, p. 57-65, Petit 1981, p. 85-89, Wood 1984, p. 45-609, Reich 1990, p. 306-318 and Hidirolou 2000, p. 21) accept de Vaux's chronological division of the pools uncritically. The fact, however, that de Vaux did not excavate and record his findings according to a sound stratigraphic system prevents us from accurately reconstructing the site's sequential building and settlement phases. As well as the problem of reconstructing a general chronological development, the archaeological evaluation of water installations presents additional difficulties. As with most water installations cut into the ground surface, it is extremely difficult to establish the exact dates of the Qumran pools. Unlike most independent structures, no foundation trenches or pre-building strata are available to provide us with data that can help determine the date of construction. From the fill one can only infer an approximate date when the pools went out of use.

Though de Vaux's preliminary reports and field notes allow us to reconstruct an approximate *terminus post quem* for each pool we are rather limited when trying to establish the *terminus ante quem*. In order to reconstruct the approximate date of construction for each pool: a) the relation between each pool and its immediate architectural surrounding should be examined carefully; and b) the intricate system of water channels that have clearly been built during various stages of the site's occupation should be studied in detail. The complexity of this task, however, necessitates a systematic re-examination of all the structural components of Qumran. This, however, goes beyond the limits of the current paper's framework.

At this point a fully accurate reconstruction of the site's stratigraphic development is utopian. We can, however, differentiate between a minimum of three different stages in the water system: the first pertaining to the presence of the circular pool; a second to the construction of the square and rectangular pools; and a third to modifications introduced to the square and rectangular pools. Given the continuous use of the circular pool and given the lack of Iron Age and other early pottery within its fill (Humbert and Chambon 1996, p. 326), its date can only be conjectured²⁹. The many examples of a completely different kind of pool (the square and rectangular installations) indicate a different building concept, representing most likely a later period. The main difference of this group of pools is not only indicated by a completely different shape, but mostly by the fact that a large percentage of its storage capacity is taken up by steps. The most obvious sign of the



Fig. 20 – Northern wall of locus 48.



Fig. 21 – Southern wall of locus 71.

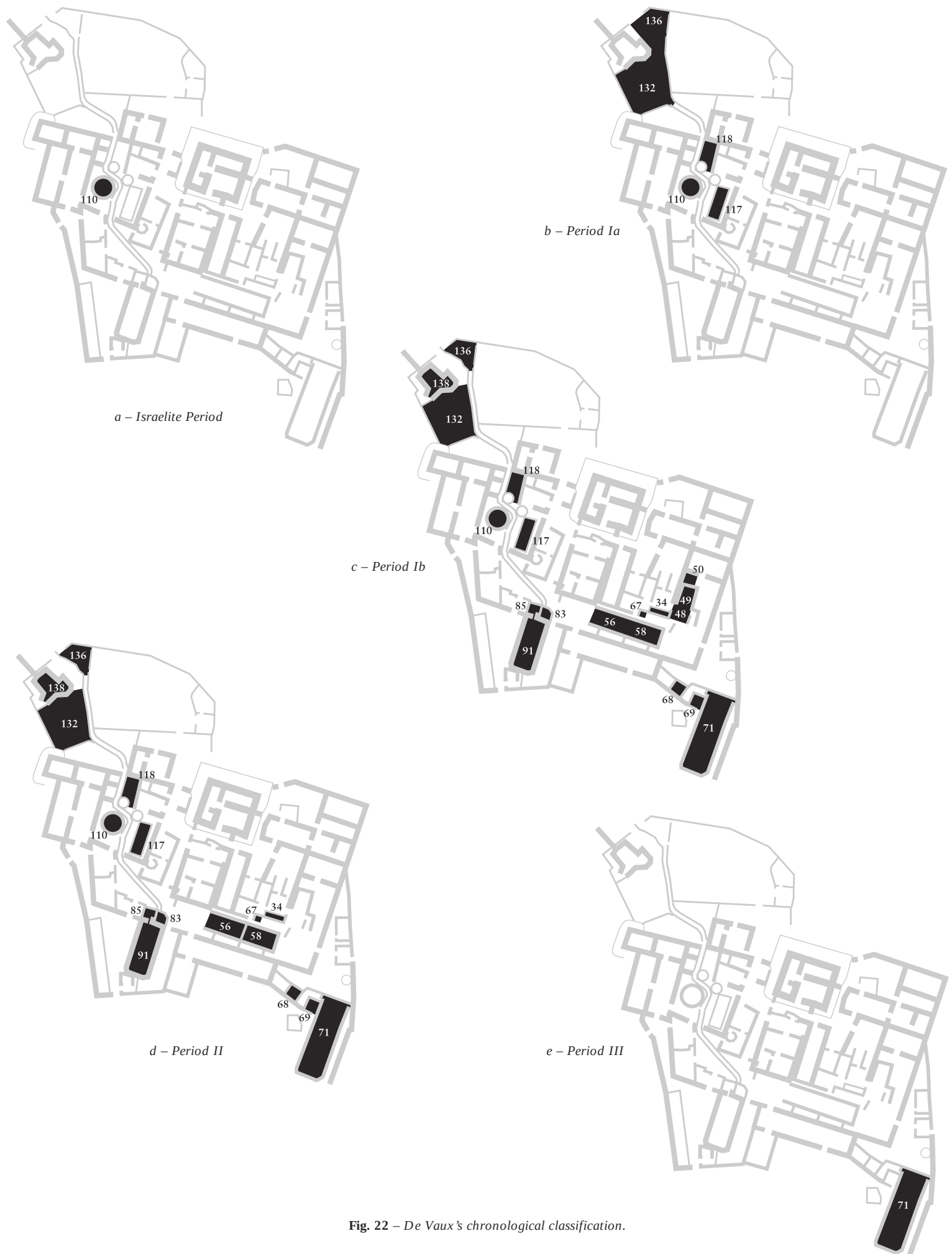


Fig. 22 – De Vaux's chronological classification.

circular pools' earlier date are its walls, whose level was heightened at some time. In an earlier stage the water entered the pool from the north. In a later stage, an additional channel, immediately connected to the newly introduced rectangular pools (loci 117 and 118) and indirectly connected to all the remaining pools at the site, entered the circular pool from the south (de Vaux 1973, p. 9 and Strobel 1972, p. 58). The clearest sign of modification introduced in this second stage is the wall separating locus 56 from locus 58. Although de Vaux does not seem to consider locus 85 as a separate pool from locus 91 (1973, p. 9), other scholars have treated them as two distinct entities (i.e. Strobel 1972, p. 60-61 and Wood 1984, p. 50). According to de Vaux, locus 85 furnishes the steps of approach to locus 91; according to others, locus 91 constitutes the only rectangular installation built from the outset as a large stepless pool. The continuous plaster application along the western walls of loci 91 and 85 strongly suggests that de Vaux's conclusion was correct. Loci 91 and 85 were most likely built as one unit. Whether the steps of locus 85 originally descended towards the bottom of locus 91 must remain speculative. An answer could possibly be found by excavating the fill in the northern part of locus 91. Given the later separation of loci 56 and 58 into a stepped and stepless section, it is likely that another separation, between a stepped part (locus 85) and a stepless part (locus 91) also took place at a later stage than that of the original construction; perhaps the two amendments were done at the same time.

Stepped and plastered pools at other Palestinian sites

De Vaux's uncertainty regarding the interpretation of the stepped pools, and his hesitation to determine their exact functions, were obviously related to the scarcity of comparative archaeological material. A comparison with similar pools excavated in Bethany (1961, p. 99), Jerusalem (Tomb of the Kings) and Samaria (1956, p. 539) convinced de Vaux that most of the pools at Qumran were not used for the purpose of ritual immersion³⁰. North, who examined these same installations more carefully (1962, p. 116-122), came to a different conclusion. Although aware of the differences regarding the typological, structural and chronological features, he believed in the pools' "cultic use" (1962, p. 122-132). One of the more vehement arguments against the view that the Qumran pools were used for ritual purity was expressed by Cross (1958, p. 67-68). According to him "the pools are typical examples of water reservoirs well known from other sites." Unfortunately, Cross' statement is not followed by a typological study of comparative archaeological material.

It was only after the 1960s that a growing number of stepped and plastered pools started to be uncovered and identified as ritual pools. The first systematic scholarly evaluation of the archaeological material (Reich 1991) was followed by a literal explosion of discoveries of "*miqva'ot*." In the early 1990s some 306 stepped pools were catalogued by Reich (1990, p. 47-61); according to Amit³¹, the number has almost doubled. Wood (1984, p. 51) and Reich (1999, p. 125), who also accept the interpretation of the Qumran pools as *miqva'ot*, claim that their features are well known from other sites. Indeed, endless examples do fit the flexible typology of the small and average-size stepped pools at Qumran³². In contrast, the large pools

constituting the majority of the plastered installations are unique to Qumran³³.

More important, though, than finding exact typological matches to some of the individual pools at Qumran, is to compare the global picture of these installations to other comparative accumulations of stepped pools³⁴. Concentrations of stepped pools were excavated in the Ophel³⁵ and the Jewish Quarter³⁶ in Jerusalem; within the context of the Dead Sea area at Masada³⁷ and Jericho³⁸; and finally at Sepphoris³⁹.

Regarding the contextual and chronological nature of the above-mentioned concentrations, three observations can be made.

1. Until recently the tendency was to view the physical act of ritual purification as being closely related to other Temple rituals, mostly performed by the priestly population. The physical proximity of the stepped pools in the Ophel and the Jewish Quarter obviously favored this concept. In his work on "Second Temple Period *Miqva'ot*," Reich explicitly relates the material phenomenon of the stepped pools, both contextually and chronologically, to the presence of the Temple. The repeated pattern of concentrations of stepped pools outside and even far from the Jerusalem context suggests this view to be overrated⁴⁰.
2. The earliest stepped pools discovered date to the Hasmonean period (second century B.C.) and appear to precede the earliest synagogues by about 100 years (Amit 1994, p. 157-158). Unlike the synagogue, which only started to flourish as an institution and only began to develop its own architectural signature after 70 A.D., the installations identified as *miqva'ot* already existed prior to the destruction of the Temple⁴¹. Stepped pools and, more importantly, concentrations of stepped pools can be found in all geographical areas of Palestine inhabited by Jews prior to and after 70 A.D.⁴²
3. Although sometimes associated with structures of a public or religious function such as the Temple in Jerusalem (Reich 1989, p. 63-65) and synagogues (Reich 1995, p. 289-297), stepped pools are more frequently found within domestic contexts⁴³.

The implications of these observations for the case of Qumran are that chronologically and contextually speaking, the existence of the many stepped pools is not unusual in itself. What does make the case of Qumran unique is the fact that the stepped pools constitute almost the only type of plastered installation.

Before the modifications that turned loci 58 and 91 into stepless pools were executed, approximately 90 percent of the total volume of water was contained in stepped pools. After this change the total volume of water contained in the stepped installations was reduced to 60 percent. Thus, at all times, most of the water at Qumran was stored in stepped pools. When compared to the above-mentioned sites that have similar high concentrations of stepped installations, the ratio is reversed. There, most of the water is stored in stepless installations.

In the houses of the Jewish Quarter in Jerusalem, the stepped pools are found in the basement levels containing other plastered installations such as bathtubs and cisterns. The cisterns are mostly significantly larger in size than the

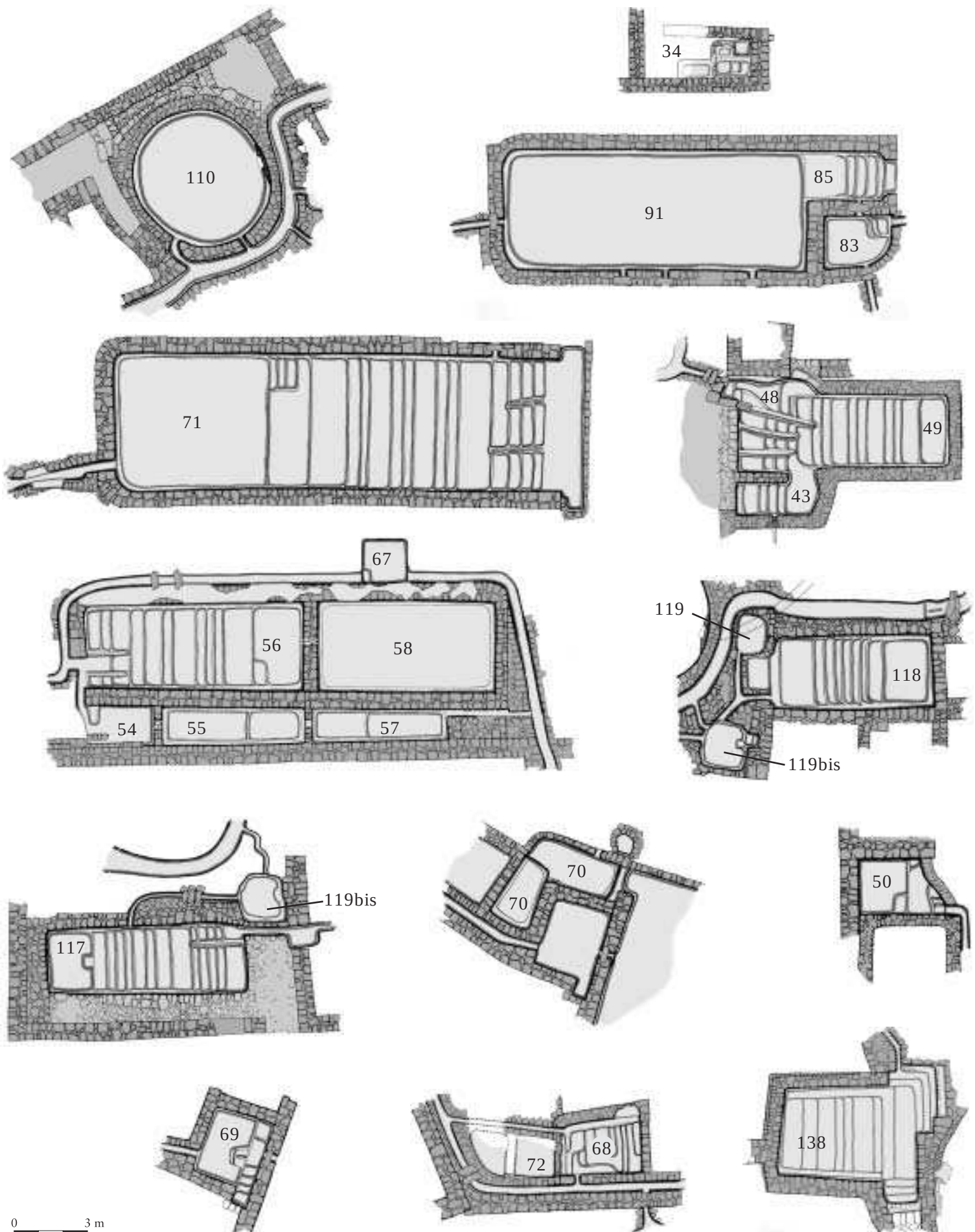


Fig. 23 – Size comparison of the plastered water installations.

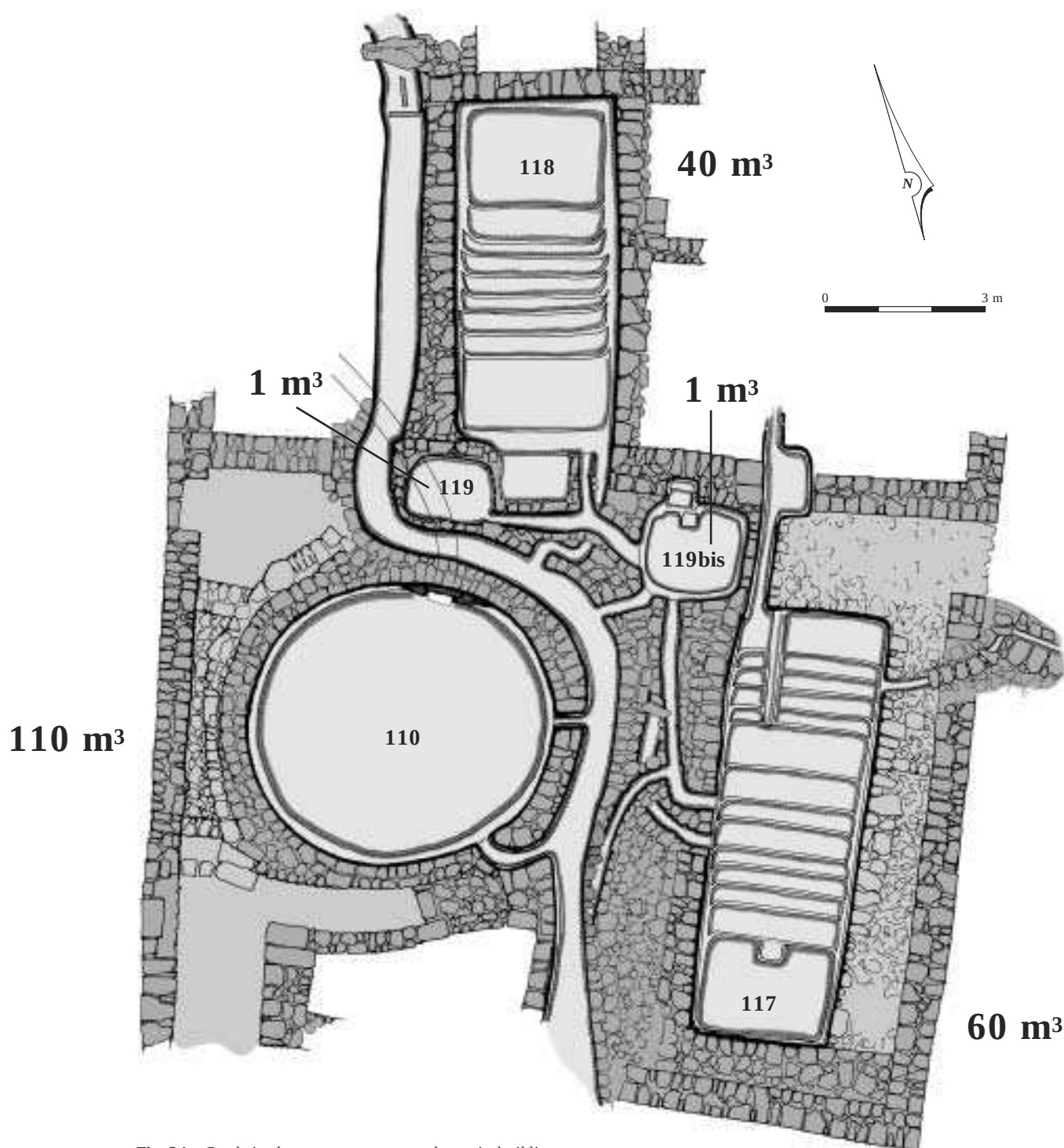
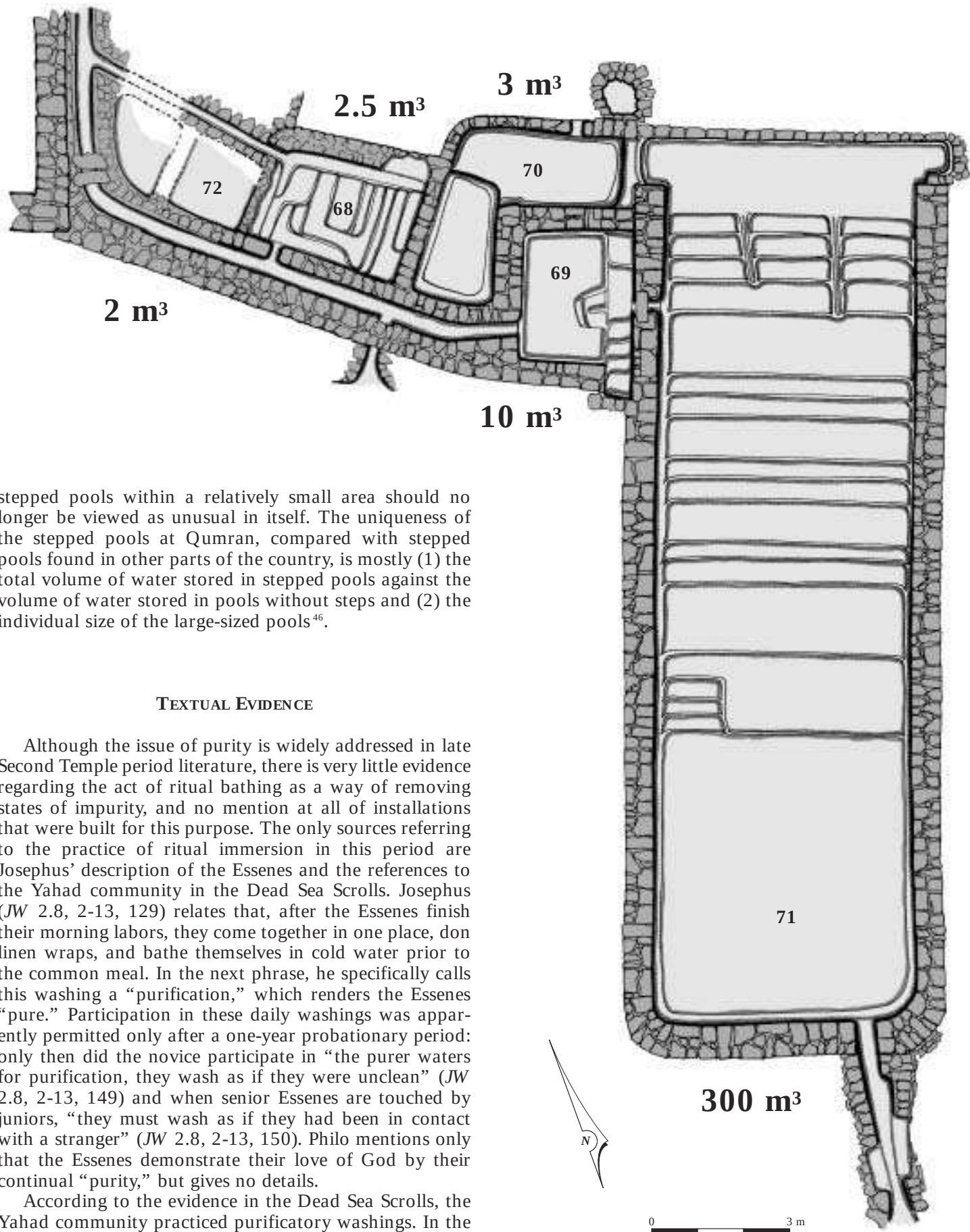


Fig. 24 – Pools in the western annex to the main building.

stepped pools and also outnumber them (Avigad 1980, p. 139-143; Reich 1990, p. 171-206; and Geva 2000, p. 29, 88, 96 and 97). The number of stepped pools at Masada is far smaller than the number of cisterns; the volume of water contained in these pools is negligible compared to the volume contained in the cisterns⁴⁴. In the Hasmonaean and Herodian palaces at Jericho, the stepped pools also appear side by side with other plastered installations. In addition to the bathtubs, cisterns and industrial installations, swimming pools can be counted among these installations (Netzer 1982, p. 11-39; 1997, p. 5-31; and 2001a, p. 333-336). In the domestic quarters of the Sepphoris Acropolis (Meyers 2000, p. 42-49; Galor and Miller), the stepped pools are usually located in proximity

to cisterns. This proximity seems to indicate that they were part of the same domestic unit. In addition to the private cisterns, several public cisterns and pools existed, considerably supplementing the storage capacity in stepless pools (Tsuk 1985, p. 40-42; 1996, p. 45-49; and 2000, p. 35-46). The Ophel in Jerusalem is the only example in which concentrations of stepped pools are found within a public setting (Reich 1990, p. 218-230 and 1998, p. 63-65). Here, too, enormous cisterns and pools are found within the same area⁴⁵.

From an archaeological point of view, the additional discoveries of concentrations of stepped pools have placed the Qumran pools into a more comprehensive context. The fact that we are dealing with a large number of



stepped pools within a relatively small area should no longer be viewed as unusual in itself. The uniqueness of the stepped pools at Qumran, compared with stepped pools found in other parts of the country, is mostly (1) the total volume of water stored in stepped pools against the volume of water stored in pools without steps and (2) the individual size of the large-sized pools⁴⁶.

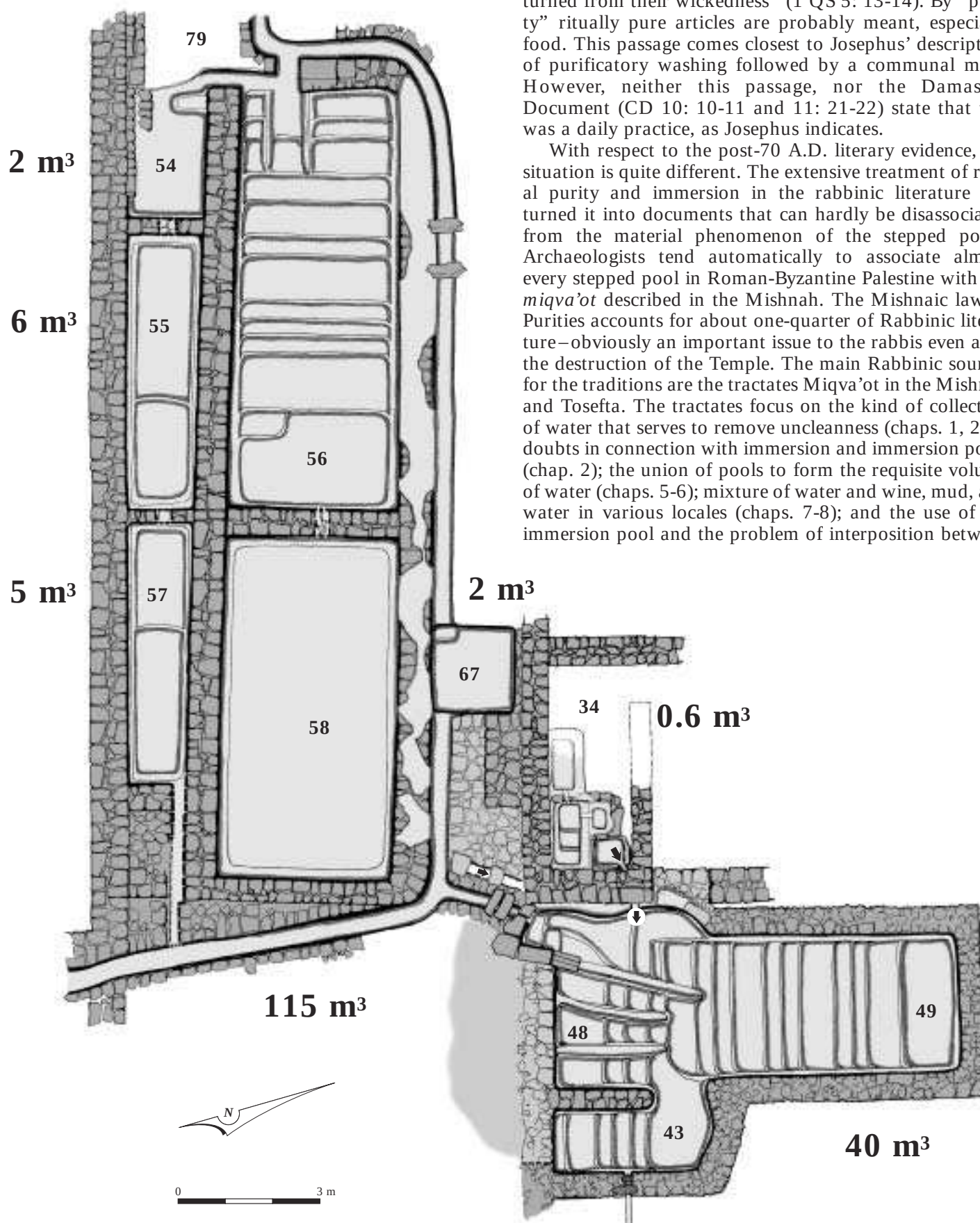
TEXTUAL EVIDENCE

Although the issue of purity is widely addressed in late Second Temple period literature, there is very little evidence regarding the act of ritual bathing as a way of removing states of impurity, and no mention at all of installations that were built for this purpose. The only sources referring to the practice of ritual immersion in this period are Josephus' description of the Essenes and the references to the Yahad community in the Dead Sea Scrolls. Josephus (*JW* 2.8, 2-13, 129) relates that, after the Essenes finish their morning labors, they come together in one place, don linen wraps, and bathe themselves in cold water prior to the common meal. In the next phrase, he specifically calls this washing a "purification," which renders the Essenes "pure." Participation in these daily washings was apparently permitted only after a one-year probationary period: only then did the novice participate in "the purer waters for purification, they wash as if they were unclean" (*JW* 2.8, 2-13, 149) and when senior Essenes are touched by juniors, "they must wash as if they had been in contact with a stranger" (*JW* 2.8, 2-13, 150). Philo mentions only that the Essenes demonstrate their love of God by their continual "purity," but gives no details.

According to the evidence in the Dead Sea Scrolls, the Yahad community practiced purificatory washings. In the *Manual of Discipline* (1 QS 3: 3-5), it is said that a man who scorns God's ordinances "shall not be absolved by expiation, or purified by lustral waters, or sanctified by seas and rivers, or cleansed by all the waters of washing"; yet, the one who humbles himself towards God's precepts

Fig. 25 – Pools from the south-east annex.

Fig. 26 – Pools in the south-eastern quadrant of the main building.



will “be cleansed when sprinkled with lustral water and sanctified in flowing water.” There does, however, seem to be a reference to some sort of purifying bath in another passage, in which one who is not obedient to God’s precepts is not to “enter the water to touch the purity of the holy men, for they will not be cleansed unless they have turned from their wickedness” (1 QS 5: 13-14). By “purity” ritually pure articles are probably meant, especially food. This passage comes closest to Josephus’ description of purificatory washing followed by a communal meal. However, neither this passage, nor the Damascus Document (CD 10: 10-11 and 11: 21-22) state that this was a daily practice, as Josephus indicates.

With respect to the post-70 A.D. literary evidence, the situation is quite different. The extensive treatment of ritual purity and immersion in the rabbinic literature has turned it into documents that can hardly be disassociated from the material phenomenon of the stepped pools. Archaeologists tend automatically to associate almost every stepped pool in Roman-Byzantine Palestine with the *miqva’ot* described in the Mishnah. The Mishnaic law of Purities accounts for about one-quarter of Rabbinic literature—obviously an important issue to the rabbis even after the destruction of the Temple. The main Rabbinic sources for the traditions are the tractates *Miqva’ot* in the Mishnah and *Tosefta*. The tractates focus on the kind of collection of water that serves to remove uncleanness (chaps. 1, 2-5); doubts in connection with immersion and immersion pools (chap. 2); the union of pools to form the requisite volume of water (chaps. 5-6); mixture of water and wine, mud, and water in various locales (chaps. 7-8); and the use of the immersion pool and the problem of interposition between

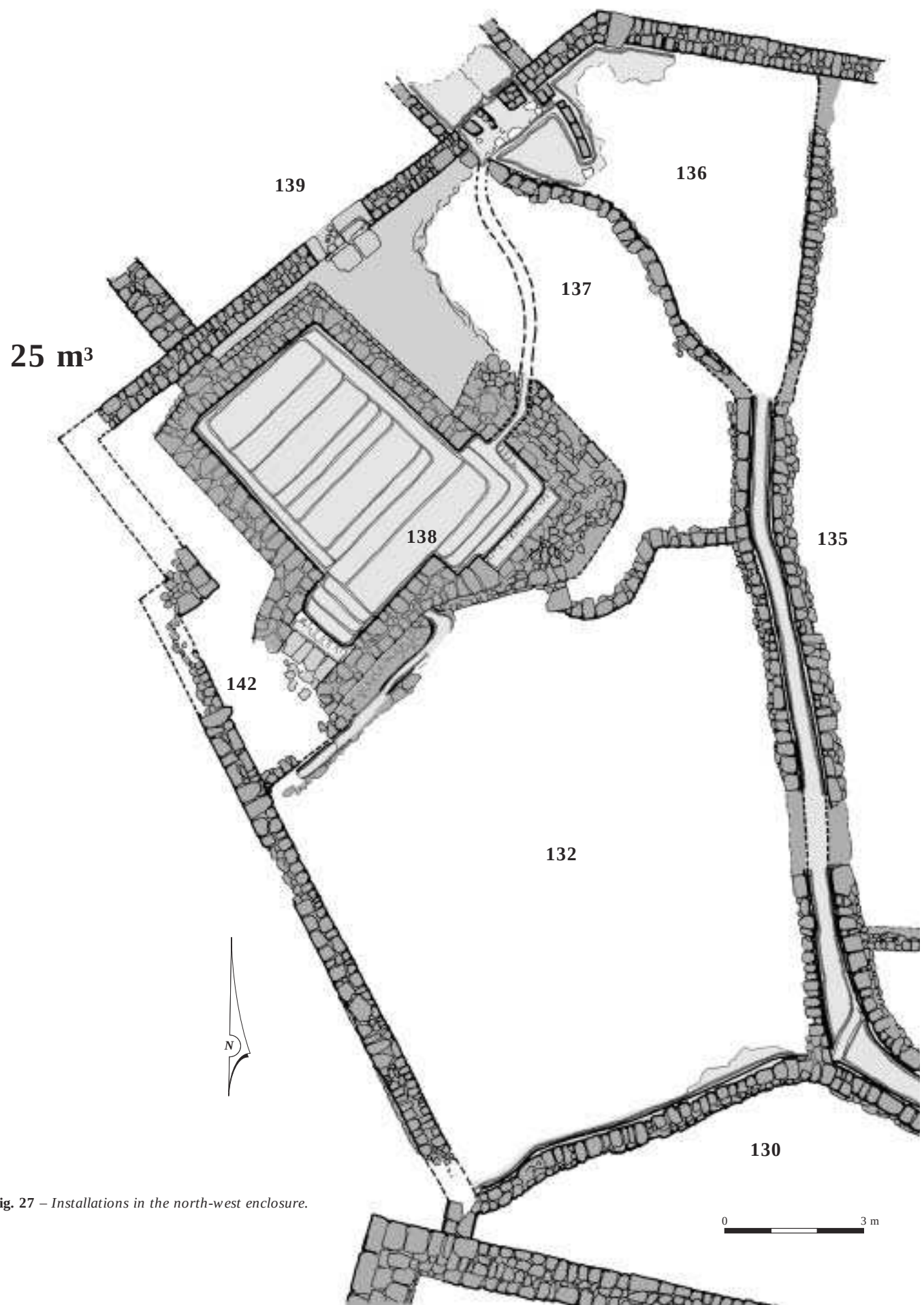


Fig. 27 – Installations in the north-west enclosure.

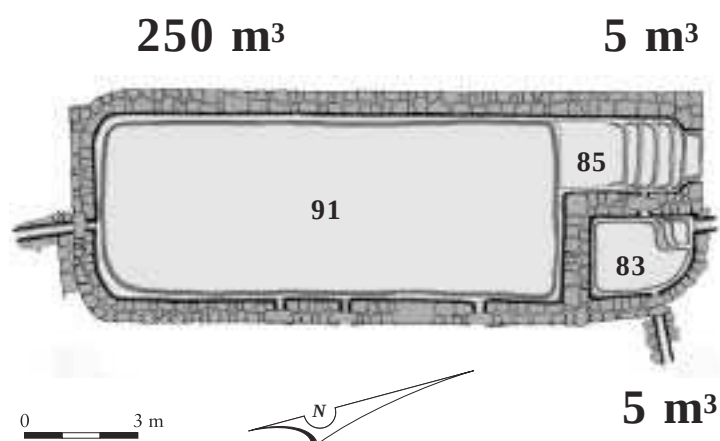


Fig. 28 – Pools in the south-west annex.

the flesh and the water (chaps. 8-10). This literature has set the grounds and interpretative framework for Reich's (1990, p. 47-62) typological study of the stepped pools. Despite the minute descriptions appearing in Rabbinic literature and its claim of treating the subject from a legalistic and rather objective point of view, the exclusive use of post-70 A.D. textual evidence for the interpretation of pre-70 A.D. material evidence seems problematic⁴⁷.

We can conclude that the stepped pools at Qumran were built at a time during which purity and ritual immersion were of great concern, in particular to the Essene and/or Dead Sea Scroll sect. Numerous rules with respect to the act of ritual immersion existed and were formulated. However, no clear prescriptions existed with respect to the features and qualities that one should associate with installations built for this purpose.

FUNCTIONS OF THE POOLS

Interpretations of the functions of the pools at Qumran are diverse. De Vaux himself changed his interpretation at least twice during the years he devoted to this project. In his excavation report he stated that "probably they were simple cisterns." Concerning two smaller installations, he concluded (1973, p. 132): "They were certainly baths, but archaeology is powerless to determine whether the baths taken in them had a ritual significance." In his last statement on the subject, de Vaux (1978, p. 983) modified his position somewhat: "This system was designed to fill the needs of a large community living in an arid region. However, the care taken in constructing these installations may suggest that they were intended for the ceremony of ritual immersion." Based on purely hypothetical grounds, Wood (1984, p. 53-58) computed the amount of water needed for normal living and bathing and compared this with the amount of water available at Qumran. He concluded that: "More than twice the required amount [of water] was available," and therefore he believed it was "the religious beliefs of the community who built and used the system" that determined its features and size. Stegemann (1998, p. 34-55), who speculatively reconstructed a fine-leather tannery within the midst of Qumran's Period Ib, considered the aqueduct supplying the large number of cisterns and water pools as mainly

connected to this industry. Locating a number of pools only vaguely somewhere within the "Main Complex," he referred to "immersion basins that were reserved for ritual cleansing...in which water had to flow in and out continuously, in order to express the symbolic force of the rite of immersion." The most recent evaluations of the pools by Reich and Hidioglou highlight once more the scholarly disagreement regarding the functions of the pools. According to Reich (2000, p. 731) "the stepped water installations at Qumran are indeed purification facilities, following a Jerusalem architectural tradition." In contrast, Hidioglou (2000, p. 19-47) strongly opposes the purifying uses of the site's pools.

Regardless of the diverse methodological approaches applied to examine and interpret the pools, only two functions are repeatedly suggested as possibilities. According to some, these installations were primarily used for the ritual act of immersion, while, according to others, they mostly functioned as cisterns.

Regarding their use as a ritual pool, from the standpoint of Jewish law, every single pool at Qumran could have functioned as a *miqveh*. First, all the pools were filled with water by exactly the same mechanism, which clearly qualifies as fulfilling the halakhic requirements of a *miqveh*. This mechanism is known as the process of *ham-sakha*, or "conduction." This mechanism allows for the pouring of drawn water onto the ground or into a channel attached to the ground from where it flows into the pool. The water thereby loses its "drawn" status⁴⁸. Second, each pool has a capacity greater than 40 *seahs* of water, which is certainly above the minimum required amount of water, corresponding to at least 250 liters. In practice, this means that even when the average and large-size pools were left with a minimal amount of water (that is, no less than 40 *seahs*), unlimited amounts of drawn water could be added from another source or reservoir. Third, except for the tubs in locus 34, no channel that could have functioned as an overflow outlet was identified at the site. According to halakhic rulings, drains and overflow channels would disqualify as installations for the function of ritual immersion. As demonstrated above, in the first stage, steps accessed all of the large pools. It should be stressed, however, that religious law does not require this element. Instead, it should be understood purely as a matter of convenience.

Regarding the use of the pools as cisterns, matters are obviously far less complicated. Halakhic considerations are not an issue. Despite the typological differences of the pools, in terms of shape, size and accessibility, each pool could have been used for storing water. The geographical isolation of the site and the difficult climatic conditions make the maximizing of water storage an important factor. Since drinking a certain quantity of water daily is not dictated by religious law but only by human physiology, the storage of drinking water must have been important.

We have demonstrated that the attribution of particular functions to a certain kind of pool has led to major discrepancies between the various interpretations of the pools' functions. It is advisable, therefore, to refrain from making clear distinctions between the pools that were used as cisterns, ritual baths and installations for other domestic purposes. The use and functions, and consequently the appearance of these installations, may have been much more flexible than generally assumed. It is very likely that,

at the outset, the stepped pools at Qumran were built and planned to fulfill a dual function, to serve as ritual immersion pools and also as cisterns. The separation of locus 56 from locus 58 and locus 91 from its stepped part, introduced some time after the initial planning and construction stage of the site, can be viewed as an attempt to eliminate certain problems that are associated with the multiple uses of the pools. These observations made in the field can be supported by a number of literary references.

From a halakhic point of view, the notion of ritually pure and impure water cannot be equated with the notion of clean and dirty water. In *Miqva'ot* 4:3 dirt fell into an immersion pool and it was pressed tight. Since the pool corresponds to all the necessary requirements and the water reached it without having been drawn, it qualifies as being ritually fit. As a matter of fact, the presence of substantive amounts of mud alone (*Miqva'ot* 2: 10) is not sufficient to disqualify a pool for use in ritual immersion.

We are also informed that, in spite of the priority given to clean water for the purpose of ritual immersion and also for drinking, this preference could not always be satisfied. In the case of drinking water, the Talmud (2: 5) differentiates between "good water," in the sense of being clear and tasty, and "bad water," that is, brackish water. The tasty water was obviously preferred over the brackish water, but people would sometimes have no choice but to drink poorly tasting and stinking water.

It is also apparent that pools used for ritual immersion and bathing and those used for the storage of drinking water were not necessarily two distinct architectural elements. It cannot be stressed enough that Talmudic references (see the etymological clarifications above), clearly state that the majority of attributes and rules that apply to ritual immersion pools also apply to other bodies of water contained in natural depressions or containers or, alternatively, built installations such as cisterns and wells. In the Damascus Document (CD 11: 1-2), a rule concerning the transportation of drinking water on a Sabbath clearly illustrates the flexibility of the use of bathing water: "On the road, if he goes down to bathe, he should drink where he stands. But he is not to draw it with any vessel." The Rabbinic recommendation of drinking the water in which one bathes is revealed in the Tannaite tradition (bSabb 41a) stating that: "He who bathes in hot water and does not drink of it, resembles an oven that is hot on the outside but not within ..." ⁴⁹ These literary references seem to clarify why making a clear-cut material distinction between cisterns and immersion pools may not always be possible.

CONCLUDING REMARKS

Fifty years ago de Vaux stated that archaeology was unable to determine the exact functions of the pools. Conventional archaeological methodology (see archaeometric contributions concerning this matter) is still powerless in determining the amount of human activity that took place inside these installations. However, the rapidly increasing amount of comparative data from contemporary sites is a valuable addition to the information available at the time of Qumran's discovery. Stepped pools are frequently found in association with structures that

can only be visited or entered after preliminary ritual washing or immersion. Stepped installations have been revealed in proximity to the Jerusalem Temple as well as adjacent to several synagogues and vine presses. In addition to Qumran, concentrations of stepped pools have been found at several other Palestinian sites (Jerusalem, Jericho, Sepphoris and Susiya). The predominantly Jewish identity of the inhabitants at the latter sites can be established both on textual and archaeological grounds ⁵⁰. The absence of concentrations of stepped pools from sites that were not inhabited by Jews is equally meaningful ⁵¹. Thus, the archaeological record clearly supports the hypothesis that stepped pools from late Hellenistic through early Byzantine times were purposely built in conformity with the halakhic requirements of *miqva'ot*. In addition to the noted similarities between the stepped pools of Qumran and stepped pools found at other sites, their differences deserve special mention. The plastered basins and pools at Qumran are much more sizeable than the surviving stepped installations known to us from other places. Furthermore, there are considerably more stepped pools than other types of plastered installations, whereas at other sites the stepped pools only constitute a small percentage of the total corpus of plastered installations. The archaeological evidence, as well as the literary evidence, examined in this chapter suggest that the dividing line between pools that were used for ritual immersion and those that were used for water storage was not as clear as previously assumed and may sometimes not be applicable. It is likely that the distinction between installations used exclusively for ritual purposes and those used solely for storage and drinking only developed over time.

Although we can clearly determine that the inhabitants of Qumran adhered to the rules of ritual purity as practiced by all Jewish sects at the time, we cannot determine their specific religious orientation or affiliation. Any expectation that the pools can solve the puzzle of "sectarian identity" is overly ambitious and non-realistic. The uniqueness of the pools at Qumran should not be over-rated.

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NOTES

- 1 According to the definition in the Oxford Encyclopedia of Archaeology in the Near East, TSUK 1997a: 350-351, this term fits best the general character of the installations found at Qumran.
- 2 Yadin asked Rabbi Muntzberg to confirm, from a Talmudic perspective, his conviction that he had excavated a ritual bath, YADIN 1966: 164-167.
- 3 Miller (forthcoming) criticizes the frequently “monolithic” use of the term *miqveh* and stresses the diverse meanings of the word, not only in Biblical, but even in later Talmudic contexts.
- 4 There is a gap in the archaeological data for both Israel and the Diaspora starting with the Early Moslem period in the mid-seventh century A.D. till around 1200 A.D. Künzl mentions the factors that have contributed to eradicate some of the archaeological evidence, KÜNZL 1992: 1-21.
- 5 According to my measurements the entire complex covers an area of 1290 square meters, and the pools, basins and channels 176 square meters. The resulting percentage differs slightly from estimated calculation, REICH 1999: 125, according to which the water installations at Qumran cover 17 percent of the total built up area.
- 6 The deviations from otherwise cited volume estimates, PETIT 1981: 87 and 94; WOOD 1984: 57-58; CANSDALE 1995: 126, stem probably from the fact that they all go back to the same source: LAPERROUSAZ’s 1976: 107-109 volume count is rather approximate, most likely a result of calculations made without existing cross-sections. NORTH’s 1962: 108 volume calculations do not seem to be cited by anyone.
- 7 The aqueduct has been repeatedly investigated, first by MASTERMAN 1903: 264-267, then after de Vaux’ discoveries by SCHULZ 1960: 50-72, and more recently by ILAN and AMIT 1989: 283-288 and 2002: 380-386. Humbert was the first to mention and describe the dam in detail, HUMBERT and CHAMBON 1994: 192 and 342.
- 8 Loci numbers 54, 55 and 57 correspond to four spaces of which the easternmost was not surely plastered.
- 9 According to geologist Aryeh Shimron, the fault in loci 48/49 could be the result of gravitational collapse within the very unstable marly terrace of Qumran. If the split in the northern wall of locus 49 were the result of an earthquake, the direction should have been towards the east, rather than towards the west.
- 10 I am grateful to Alain Chambon who indicated to me the importance of those basins and called my attention to their original state of preservation as recorded in the photographs. See HUMBERT and CHAMBON 1996: 43 and 66.
- 11 Those observations were made by J.-B. Humbert.
- 12 One example would be the channel opening in the southwestern corner of locus 138. According to de Vaux, this was the location of the overflow channel, and according to Strobel, it was the supply channel of the pool.
- 13 Numerous contemporary examples of impeccably preserved vaulted ceilings covering pools and cisterns can be found throughout the country, such as these at Sepphoris, HÖGLUND and MEYERS 1996: 39-43; and Chorazin, YEIVIN 1962: 153 in the north, and in the Ophel, REICH 1990: 87-93 and 1998: 63-65 and Jewish Quarter excavations in Jerusalem, AVIGAD 1983: 139-43 and GEVA 2000: 29, 88, 96 and 97.
- 14 The annual evaporation loss at Qumran is about 2,1 m per year, WOOD 1984: 57. By covering an open surface, however, the amount of evaporation is significantly reduced, KAZMANN 1972: 64.
- 15 It is J.-B. Humbert who suggested that some of the rooms might have been covered with palm tree trunks. He also informed me that the maximum length of a grown date palm is 9 m. The diameter of the circular pool (locus 110) is 5,10 m and the maximum width of the largest rectangular installation (locus 91) is 4,82 m, therefore not exceeding the maximum width of other architectural spaces at the site.
- 16 The earliest evidence of the date palm, *Phoenix dactylifera*, at Tel al-Oueili and Ubaid Eridu, suggests that southern Mesopotamia was the center of domestication of the date palm. See HUOT 1987: 243.
- 17 It has been suggested that the “open cisterns” of the Negev mountains were covered with hides to reduce evaporation. See TSUK 1997: 13.
- 18 For alternative roofing methods at Masada, see NETZER 1991: 613.
- 19 See in particular Miqva’ot 6: 1, where it is clearly stated that: “Any pool of water which is mingled with [water of] an immersion-pool is [deemed to be as valid] as the immersion pool.” Translation NEUSNER 1976: 38.
- 20 PETIT 1981: 94 points out the problems associated with estimations of daily water consumptions. Within certain areas of modern Lebanon, the daily water consumption per person can be as low as three liters. In the US, an individual can use up to 363 liters per day. In ancient Rome the daily water consumption per inhabitant could reach up to 1000 liters.
- 21 Scholars no longer attempt – as in previous generations – to equate units of volume mentioned in the Bible with Greek and Roman measures. The only method by which modern scholars can determine the values of dry and liquid measures is to measure the volume of vessels discovered in excavations in Palestine whose capacity is marked on them. According to W.F. Albright’s calculations, which are accepted by most scholars today, the “royal bath” (*bt lmlk*) has a capacity of 22 liters. Accordingly, one *seah* corresponds to 7,3 liters and 40 *seahs* to 292 liters.
- 22 REICH 1980: 225-256 was the first to elaborate the theory that the divided stairways were intended to separate those who had already bathed from those who were still impure. For other pools with partitioned stairs see AMIT 1997: 35-48 and 1999: 75-84.
- 23 Josephus mentions that the Essenes are divided into four groups “according to the duration of their training” (*JW* 2.8, 10, 150-153). MICHEL and BAUERNEFELD 1960: 437, believe that the four groups are the priests, levites, laity, and novices. PELLETIER 1975: 36, on the other hand, regards the four classes as children, first-stage novices, second-stage novices, and full members. Finally, THACKERAY 1967: 381, and many others regard the four classes as the first-year novices, second-year novices, third-year novices, and full members.
- 24 In addition to examining the entire photo collection of de Vaux at the École Biblique, I also examined early pictures taken by other scholars. NORTH’s 1962: 115 plate III, study of the Qumran installations includes a depiction of locus 138 that indicates the outstanding state of preservation during the years immediately following the excavation.
- 25 At Qumran most of the building stone is the local limestone, although a small percentage consists of dolomite. I would like to thank Aryeh Shimron for explaining to me that only a simple chemical test with hydrochloric acid will tell if the material is limestone or dolomite. This difference cannot usually be detected with the naked eye.
- 26 Shimron’s chemical analysis of the plaster of the pools determined that each group of construction types falls into a different range of cadmium composition.
- 27 In his final report on The Building Stratigraphy and Architecture at Masada, NETZER 1990: 607-613 describes the various masonry styles. Though in the subtitle he refers to the “various Herodian building phases” he does not specify the chronological sequence of the different styles.
- 28 Suggested dates for the beginning of the Hellenistic settlement range between the mid-second century B.C. and ca. 100 B.C. or even later. The 30-plus-year occupational gap postulated by de Vaux between Periods Ib and II has been reduced to a few years by some, or completely eliminated by others. Those chronological questions have been examined in depth by a number of scholars. Magness has made some tentative suggestions regarding the ceramic evidence (1994: 39-49 and 1995: 58-65). For the numismatic evidence, see MESHORER 1987: 158. More recently, a very useful discussion of chronological issues with relevant references was written by BROSHI 1992: 111.
- 29 Although de Vaux and others have suggested that circular pools have been found at a number of Iron Age sites in Palestine, this claim cannot be substantiated. In spite of the cylindrical shape of the pools excavated at Gibeon, PRITCHARD 1961: 7-12 and fig. 26, the spiral staircase gives it a rather different character. For the most common types of water reservoirs see RUBIN 1988: 229-244; SHILOH 1992: 275-293 and TSUK 1997b: 12-13.
- 30 BENOÎT and BOISMARD 1951: 200-206, who excavated the installations at Bethany, excluded the possibility of a ritual function.
- 31 Amit and Zissu have catalogued all ritual pools excavated after 1990. As of the writing of this paper, the catalogue has not been published.
- 32 The most complete catalogue of illustrations as of this date was put together by REICH 1990: 373-473.
- 33 The largest installation identified as a ritual pool was excavated at Alon Shvut by AMIT 1999: 75-84. This pool, however, if indeed used for the purpose of ritual immersion, was found in isolation and served large numbers of by-passing pilgrims.
- 34 No sites with less than 10 stepped pools are being considered in the present study.
- 35 The stepped pools in the area south and southwest of the Temple Mount platform are briefly described by REICH 1991: 218-230 and 1998: 63-65. However, no detailed study of the latter pools has been published to this date.
- 36 AVIGAD’s 1980: 139-143 and REICH’s 1991: 171-206, descriptions are more informative than the ones appearing in the final report, GEVA 2000: 29, 88, 96 and 97.
- 37 A total of 11 stepped pools was excavated by YADIN 1966: 164-167 and integrated within the final report written by NETZER 2001a: 13-17, 76-87, 127-129, 158, 181, 183, 221-229, 260-261, 398, 507-510.
- 38 The water systems at the Hasmonean and Herodian Palaces at Jericho are only treated briefly in the final report, NETZER 2001: 333-334. More detailed studies of the ritual pools were published previously, NETZER 1982: 106-119 and 1997: 5-31.
- 39 MEYERS, NETZER, and MEYERS 1986: 17; HÖGLUND and MEYERS 1996: 39-43; MEYERS 2000: 42-49; GALOR (forthcoming) and MILLER (forthcoming).
- 40 MILLER (forthcoming), mostly from a historical point of view, and GALOR (forthcoming), mostly from an archaeological perspective, use the Sepphoris example as a case study to stress the importance of ritual purity concerns outside of the context of the Jerusalem Temple.
- 41 The controversy on Second Temple period synagogues is mostly a result of the difficulties associated with identifying their characteristic features during this early period. See for example LEVINE 1987: 7-11.
- 42 The date of the destruction of the Jerusalem Temple is far too frequently imposed as an artificial barrier between different currents of material cultures. This is not only the case for the study of stepped pools or “*miqva’ot*” but also for the study of stone vessels. REGEV 2000: 229-237 uses examples of ritual pools and stone vessels as indicators of strict observances of ritual purity laws before 70 A.D. and the radical decline of the latter structures and objects as proof of a change after 70 A.D. The continuous existence of stepped pools during the late Roman and even Byzantine periods at Sepphoris (GALOR forthcoming) and Susiya, NEGEV 1985: 231-252, indicates continuity in the use of stepped pools. The continuous existence of stone vessels into the second century A.D., GIBSON and LASS 2000: 72, suggests a similar continuity in the material culture. SCHMIDT 1994: 37-63, criticizes the artificial use of the date 70 A.D. in a historical context.
- 43 Miller’s (forthcoming) elaborate analysis of the Rabbinic literature stresses the concerns for ritual immersion rites within the domestic context. AMIT 1996: 47 differentiates between ritual pools in urban and rural domestic contexts.
- 44 The cisterns were not included within the final report on the architecture, NETZER 1990. A brief description appears in YADIN’s 1966: 26-33, popular book.

- 45 Surprisingly the large stepped pools at Qumran (loci 56/58, 71 and 91) are about three times larger than the largest pools in the Ophel, REICH 1990: 230-243. Even if the pools at Qumran were used by all the inhabitants of the site, or alternatively, if the site was used as a cultic center in which large number of believers would assemble to pray, eat, immerse and work together, the numbers at Qumran surely would not approach the daily traffic by the Temple Mount in Jerusalem.
- 46 Several other stepped pools comparable in size to the largest Qumran pools have been uncovered at other sites. Those, however, are single pools and do not form part of a concentration of stepped installations. See AMIT 1999: 75-84 and CORBO and LOFFREDA 1981: fig.1.
- 47 WRIGHT 1995: 190-191, recognized the severity of this problem.
- 48 Miller (forthcoming) describes the differences between the Biblical, Tannaitic and Amoraic laws with respect to hamsakha.
- 49 Those two passages do not refer to ritual bathing.
- 50 AMIT 1996: 62 and 1994: 158, was able to show the connection between the presence of stepped pools and Jewish settlements.
- 51 This does not include sites where individual stepped pools were found. The stepped pool at 'Ain ez-Zâra, CLAMER 1997: 110, is an exact parallel to the pools 117 and 118 of Qumran. And it is highly probable that the inhabitants of ez-Zara – Callirhoe in the dependence of Machaerus, were Jews.

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